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Supplementary Material: Physical Gradient Recovery and Stopping Guarantees for a Folded Stokes Parametrization

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Core recovery details and validation are in Sections S4–S5, S8–S9, and S11–S15. Sequential initialization, restricted flow, and the fixed two-frame verification are in Sections S16–S17. Auxiliary stationary geometry, projection, renderer construction, and coordinate dynamics are in Sections S1–S3, S6–S7, and S10. We retain the main paper's conventions: $p = (Q, U)$, $g_p = \nabla_p f$, $H_{pp} = \nabla^2_{pp} f$, $r = \|w\|_2$, $e = w/r$ for $r > 0$, and $P_\perp = \mathrm{Id}_2 - ee^\mathsf{T}$. For $M \succ 0$, $\|x\|^2_M = x^\mathsf{T} M x$ and $\|A\|_{2\to M} = \|M^{1/2}A\|_2$. The disk-projection map is denoted $P(w)$; $P_\perp$ is the tangential matrix.

S1. COMPLETE IDEAL-METRIC CRITICAL-POINT ANALYSIS

We use the linear-Stokes cone $\mathcal{C} = \{(I, Q, U) : I \geq \sqrt{Q^2 + U^2}\}$ and $T_H(a, w) = I(a)(1, H(w))$, where I is a smooth bijective coordinate onto $(0, \infty)$ with nonzero derivative. The maps are

$$\begin{aligned} F(w) &= 2w/(1 + r^2), \quad G(w) = w/\sqrt{1 + r^2}, \\ P(w) &= w/\max\{1, r\}, \qquad r = \|w\|. \end{aligned} \tag{S1}$$

For ideal equal-gain analyzers at $0, 45, 90, 135$ degrees, observations $y \in \mathbb{R}^4$ have mean As, where

$$A = \frac{1}{2}\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}, \quad W = A^\mathsf{T}A = \mathrm{diag}(1, \tfrac{1}{2}, \tfrac{1}{2}). \tag{S2}$$

The intensity measurements and Stokes components share a unit. Equal independent analyzer variances give this metric up to a common positive scale. The factor $1/2$ is the ideal linear-analyzer transmission; it yields the two polarization weights $1/2$ in W. Our loss is

$$\ell(s; s^*) = \tfrac{1}{2}\|s - s^*\|^2_W. \tag{S3}$$

The metric is fixed throughout the analysis; a state-dependent covariance defines a different objective.

A. CLASSIFICATION WITH TRAINABLE INTENSITY

Let $s^* = I^*(1, de)$ with $I^* > 0$, $0 < d < 1$, $\|e\| = 1$, and retain W in (S2). A strict saddle here means a stationary point with at least one negative Hessian eigenvalue. It need not have a nonsingular Hessian.

Theorem S1 (Complete quadratic classification)**.** *For $\ell(T_F(a, w); s^*)$, the finite stationary points are the two exact preimages of s^* and two fold states*

$$w_\sigma = \sigma e, \qquad I_\sigma = I^* \frac{2 + \sigma d}{3}, \quad \sigma \in \{-1, 1\}. \tag{S4}$$

Both fold states are strict saddles. In local coordinates $(I, \text{radial } w, \text{tangential } w)$ their Hessian is diagonal, with entries

$$\left(\frac{3}{2}, -\frac{1}{2}I_\sigma(I_\sigma - \sigma I^* d), \frac{1}{2}I_\sigma \sigma I^* d\right). \tag{S5}$$

At the aligned fold ($\sigma = 1$),

$$\ell_b = \frac{(I^*)^2(1 - d)^2}{6}, \qquad \partial^2_{rr}\ell_b = -\frac{(I^*)^2(2 + d)(1 - d)}{9}. \tag{S6}$$

The physical output is not a constrained stationary point.

Proof. Away from $r = 1$, DF is invertible and the regular intensity coordinate makes T_F locally invertible. A stationary composite loss must then have $s = s^*$. At $r = 1$, the tangential condition forces $w = \sigma e$. The intensity condition is

$$(I - I^*) + \tfrac{1}{2}(I - \sigma I^* d) = 0,$$

which gives (S4). Differentiating in the stated local coordinates, using the radial second derivative $\phi''(1) = -1$, gives (S5); mixed entries vanish. The radial entry is negative for both signs. A regular change from I to a preserves Hessian inertia at stationarity. Substitution gives (S6).

At the aligned state the physical gradient is

$$\nabla_s \ell = \frac{I^*(1 - d)}{3}(-1, e).$$

Its negative points into the cone and strictly reduces the loss. Yet the head Jacobian kills this normal vector, giving zero raw gradient. Hence the output fails the constrained first-order condition. □

In particular, holding $I = I^*$ at the fold would miss the joint stationary point: its intensity gradient is generally nonzero. Accounting for intensity changes both the location and the classification of the critical states. These saddles are consistent with local-minimum preservation of the folded ball map [1]; a non-optimal critical point need not be a local minimum.

Proposition S2 (Monotone and projection controls)**.** *For the same interior target, $\ell \circ T_G$ has exactly one finite stationary point, which attains s^*. For T_P, the unique interior minimum attains s^*. Every exterior state $w = \sigma re$, $r > 1$, at $I = I_\sigma$ is stationary. The aligned family consists of suboptimal flat local minima; the anti-aligned family consists of saddles.*

Proof. G and the regular intensity coordinate form a diffeomorphism onto the positive-intensity cone interior. An interior stationary point of the strictly convex quadratic is therefore its exact target. For exterior P, the radial derivative is zero and the remaining stationarity equations are those above. The Hessian entries are $3/2$, 0, and $I_\sigma \sigma I^* d/(2r^2)$. For the aligned family, the loss is locally minimized in intensity and direction and is constant in raw radius. At $r = 1$, P is nonsmooth and no classical derivative is asserted. □

These examples describe the normalized disk map P inside a trainable head. Figure S1 compares its raw-coordinate landscape with the folded and monotone maps.

A nonzero cone normal also explains the extension to arbitrary smooth feasible full heads whose open raw neighborhood maps into $\mathcal{C}$: at a finite fully polarized state $s = (I, Ie)$, the normal $(-1, e)$ is annihilated by the head Jacobian. If all samplewise loss gradients have this form, the chain rule gives zero gradient for any upstream backbone realizing those outputs.

S2. CONVEX POST-HOC PROJECTION REFERENCE

We also retain the convex reference. For any $v \in \mathbb{R}^3$, let $P_W(v)$ be its W-projection onto $\mathcal{C}$. For any $s^* \in \mathcal{C}$,

$$\|P_W(v) - s^*\|_W^2 \le \|v - s^*\|_W^2 - \|v - P_W(v)\|_W^2. \tag{S7}$$

This standard projection inequality [2] applies to arbitrary network outputs as well as least-squares estimates. Separately, when v is the unconstrained least-squares estimate, $P_W(v)$ solves the ideal observation-constrained least-squares problem. For $v = (t, y)$, $\rho = \|y\|_2$, the projection is unchanged if $t \ge \rho$; otherwise, for $\rho > 0$,

$$b = \max\{0, (2t + \rho)/3\}, \qquad P_W(v) = (b, by/\rho). \tag{S8}$$

At $\rho = 0$, the answer is $(\max\{t, 0\}, 0, 0)$. The inequality measures post-hoc error in W.

S3. SAVED RENDERER AND CONSTRUCTED STATIONARY FIELD

A. REFERENCE CONFIGURATION

Figure S2 shows the saved 32×32 reference, generated with Mitsuba 3.9.1 in its documented polarized mode. The scene is a radius-one GGX rough-conductor sphere, roughness 0.15, spectral constants $\eta = 0.2$ and $k = 3$, and a uniform unit environment. A perspective camera at $(0, 0, 4)$ looks at the origin, with 35-degree field of view and box pixel filter. Four independent renders, seeds 821–824 and 1,024 samples per pixel each, were averaged.

We use the first RGB output channel of the spectral renderer and the saved Stokes basis. Intensities are relative radiance in arbitrary units. The stored fourth Stokes component is zero. Full-image mean DoLP is 1.311%, maximum DoLP 4.744%, and intensity ranges from 0.97337 to 1.21369. This weak-polarization, finite-sample array is the target used in the stationary-field construction below.

B. A CONSTRUCTED IMAGE-BASED STATIONARY COUNTEREXAMPLE

For each of the 1,024 reference pixels, retain I^*, d, e and initialize the folded head at the aligned state in (S4): $I_b = I_+ = I^*(2 + d)/3$, $s_b = I_b(1, e)$. The construction supplies the reference direction to the initialization. At $d = 0$, choose any unit direction. Set $I_0 = 1$ and $a = \log(\exp(I_b) - 1)$, so the intensity activation is the same softplus used in the theory.

Every initialized output is physically admissible and has DoLP one. Its normalized weighted error is exactly

$$E_b = \frac{\|s_b - s^*\|_W^2}{(I^*)^2} = \frac{(1 - d)^2}{3}. \tag{S9}$$

Figure S2 shows the constructed field. The mean E_b is 0.324735, with range 0.302460–$1/3$. Over 1,000 independent-pixel raw-coordinate gradient steps of size 0.1, the largest raw-gradient component is 2.66×10^{-16} and the largest parameter change is 4.44×10^{-16} in this float64 run. The mean normalized half-loss remains 0.162368.

The field has zero head gradient and admissible outputs, yet a positive objective gap. Its negative curvature permits escape under perturbations. The archive supplies the reference, initial and final coordinates, and per-step losses.

S4. FINITE-RADIUS DERIVATIVES AND ERROR ENVELOPES

All quantities in this section are evaluated at known $I > 0$, $w = re$, $r > 0$, $\|e\| = 1$, and $s_r = I(1, \phi e)$ with $\phi = 2r/(1 + r^2)$. The physical objective is the fixed quadratic $L(s) = \frac{1}{2}\|s - z\|_W^2 + c$, $W \succ 0$, and $g = \nabla L(s_r)$. The rational head has the

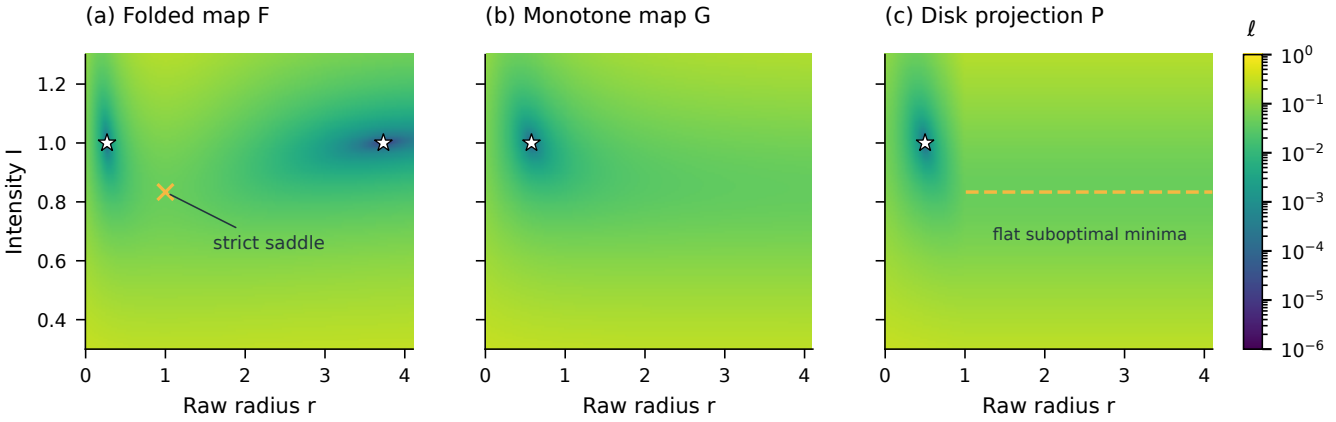


FIGURE S1. **Aligned (I, r) slices for $I^* = 1$, $d = 0.5$. Stars mark exact solutions. The folded cross is a strict saddle at $I = 5/6$, $r = 1$. The exterior disk-projection family has flat positive loss. Only the target-aligned direction is shown; the theorem also classifies anti-aligned states. A shared logarithmic scale clips values below 10^{-6}.**

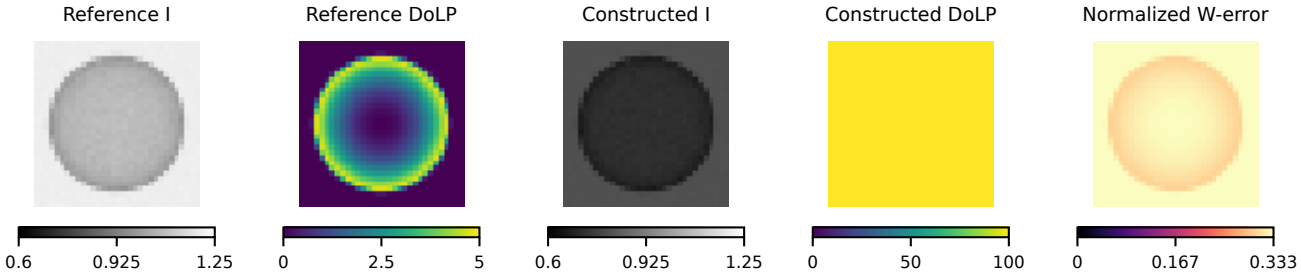


FIGURE S2. **A frozen numerical reference and a constructed stationary field on its 32×32 grid. The first two panels show the saved first-RGB-channel intensity and DoLP. The next two use the reference directions and the analytically stationary intensity; they are a mathematical counterexample field. Reference and constructed intensity share a scale. Reference DoLP spans 0–5 percent, while the constructed output has DoLP one and uses 0–100 percent. The final panel is normalized weighted error.**

exact derivatives

$$\begin{aligned} \phi &= \frac{2r}{1+r^2}, & t_r &= \frac{2}{1+r^2}, \\ a_r &= \phi' = \frac{2(1-r^2)}{(1+r^2)^2}, & \kappa_r &= \phi''(r) = \frac{4r(r^2-3)}{(1+r^2)^3}, \\ \ell_r &= t_r' = -\frac{4r}{(1+r^2)^2}, & J &= a_r e e^{\mathsf{T}} + t_r P_\perp, \\ K &= \kappa_r e e^{\mathsf{T}} + \ell_r P_\perp. \end{aligned} \tag{S10}$$

Here $J = DF(re)$ and $K = D(DF)(re)[e]$. In particular, $Je = a_r e$, $\|J\|_2 = t_r$, and $K = -\mathrm{Id}$ at $r = 1$.

A. RECOVERY BIAS AND ITS BOUND

The composite Hessian splits into physical-Hessian and map-curvature terms. Radial multiplication gives

$$\begin{aligned} v_r &= I^2 J^{\mathsf{T}} W_{pp} J e + I K g_p \\ &= I^2 a_r J W_{pp} e + I K g_p, \\ b_r &= g_I + \phi e^{\mathsf{T}} g_p. \end{aligned} \tag{S11}$$

Set $e_0 = (1,0,0)^{\mathsf{T}}$ and $S_e = (e, -\mathrm{Id}_2)^{\mathsf{T}} \in \mathbb{R}^{3\times 2}$. Using the exact-fold formula $g^{\mathrm{naive}} = e_0 b_r + S_e v_r / I$ at this radius gives the exact bias

$$\begin{aligned} g^{\mathrm{naive}} - g = (\phi - 1)(e^{\mathsf{T}} g_p) e_0 \\ + S_e \{ I a_r J W_{pp} e + (K + \mathrm{Id}) g_p \}, \end{aligned} \tag{S12}$$

Apply the $M = W^{-1}$ norm, with $k_b = \|e_0\|_M$, $k_v = \|S_e\|_{2\to M}$, and supplied bound $\|g_p\|_2 \le G_p$. Adding derivative errors $|\delta b| \le \varepsilon_b$, $\|\delta v\|_2 \le \varepsilon_v$ gives the scalar naive-reuse envelope used in the prescribed-envelope comparison:

$$\begin{aligned} \eta_r = k_b \{ (1-\phi) G_p + \varepsilon_b \} \\ + k_v \{ I |a_r| \|J\|_2 \|W_{pp}\|_2 + \|K + \mathrm{Id}\|_2 G_p + \varepsilon_v / I \}. \end{aligned} \tag{S13}$$

Its scalar coefficients are given above. The physical model in this retained bound is quadratic.

For a tighter envelope define

$$\begin{aligned} k_r &= I a_r S_e J W_{pp} e, \\ A_r &= (\phi - 1) e_0 e^{\mathsf{T}} + S_e (K + \mathrm{Id}). \end{aligned} \tag{S14}$$

Then

$$\|g^{\text{naive}} - g\|_M \le \|k_r\|_M + \sqrt{\lambda_{\max}(A_r^{\mathsf{T}} M A_r)}\, G_p. \tag{S15}$$

This refinement follows directly from (S12); the prescribed-envelope plots in Section S7 use the scalar envelope.

A known model gives one conservative choice without a fitted derivative error:

$$G_p = I\|W_{pI}\|_2 + I\phi\|W_{pp}\|_2 + \|(q_W)_p\|_2, \qquad q_W = Wz. \tag{S16}$$

This choice uses the exposed model quantity q_W. Otherwise, the application must supply a bound G_p.

Writing $\delta = r - 1$ gives $1 - \phi = \delta^2/(1 + r^2)$, $a_r = -\delta + O(\delta^2)$, $1 + \kappa_r = 3\delta + O(\delta^2)$, and $1 + \ell_r = \delta + O(\delta^2)$. The physical-Hessian bias is generally first order. For a general C^2 objective, replace W_{pp} by its local physical Hessian block; a bound on this block is additionally needed. Radius alone does not bound the recovery error.

B. WHY SUPPORT SLACK CANNOT BE OMITTED

Set $n = (-1, e)$ and $\sigma_r = I(1 - \phi) \ge 0$. Since $n^{\mathsf{T}} u \le 0$ for every $u \in \mathcal{C}$ and $n^{\mathsf{T}} s_r = -\sigma_r$, completion of the quadratic gives, for every $\lambda \ge 0$,

$$\begin{aligned} L(s_r) - L(u) &\le -(g + \lambda n)^{\mathsf{T}}(u - s_r) \\ &\quad - \tfrac{1}{2}\|u - s_r\|_W^2 + \lambda\sigma_r \\ &\le \tfrac{1}{2}\|g + \lambda n\|_M^2 + \lambda\sigma_r. \end{aligned} \tag{S17}$$

Gradient-error inflation gives the main paper's near-fold bound. Its support plane is $n^{\mathsf{T}} u \le 0$, not the plane through the interior s_r.

For the counterexample take $e = (1, 0)$, $r = 4/5$, $I = 1$, $W = \operatorname{diag}(1, 1/2, 1/2)$. Then $\phi = 40/41$ and

$$z = (4/5, 282/205, 0), \quad P_{\mathcal{C}}^W(z) = (122/123, 122/123, 0). \tag{S18}$$

The exact gradient at s_r is $-n/5$, so its distance to the negative normal ray is zero. The rational coordinates above are chosen to give this gradient while keeping s_r interior; direct substitution gives the positive gap $241/50430$. The corrected half-space minimizer is feasible and attains the bound. The executable independent check retains the analytic expressions, sequential least-squares programming (SLSQP) comparison, and floating-point tolerances.

S5. NUMERICAL AUDIT DEFINITIONS AND ACCESS MODEL

For a candidate cone quadratic-program (QP) solution x, write $h(x) = x_I - \|x_p\|_2$ and $g_x = W(x - z)$. In the 84 stationary and 108 nonstationary retained checks the solver starts at $(1.5, 0, 0)$, uses analytic first derivatives, SLSQP objective tolerance 10^{-13}, and at most 1,000 iterations. On the non-apex boundary, define $n_x = (-1, x_p/\|x_p\|_2)$ and the multiplier and Karush–Kuhn–Tucker (KKT) residuals by

$$\begin{aligned} \lambda_x &= \max\{0, -g_x^{\mathsf{T}} n_x/\|n_x\|_2^2\}, \\ R_{\text{stat}} &= \|g_x + \lambda_x n_x\|_\infty, \\ R_{\text{comp}} &= |\lambda_x h(x)|, \qquad R_{\text{feas}} = \max\{0, -h(x)\}. \end{aligned} \tag{S19}$$

The retained implementation uses 10^{-7} as its numerical boundary-classification tolerance: it classifies $h(x) \le 10^{-7}$ as boundary and otherwise sets the multiplier to zero. The smallest solution norms in the 84 and 108 retained cases are 1.452 and 0.4142, respectively. All 84/108-case solutions are away from the cone apex; the sensitivity implementation checks the apex separately. At the apex the full normal cone must be used: optimality is $g_0 \in \mathcal{C}$, rather than a single fixed normal. Euclidean projection onto $\mathcal{C}$ gives a cone-distance residual there. The single-normal residual above applies away from the apex.

The constructed derivative-only wrapper exposes composite value, head first derivatives, normalized intensity derivative, radial HVP, and the fixed metric, while its verification harness retains a physical-gradient oracle. The harness compares recovered and direct gradients; the restricted interface is an implementation of a stipulated information model. Sensitivity perturbations use deterministic envelopes fixed before their errors are evaluated. Finite-difference HVP checks use central differences of the analytic raw gradient; their observed discrepancies test the implementation but are not universal error envelopes.

For the 82-case sensitivity block, the SLSQP starting point is $(1.5, 0.1, -0.1)$ and the iteration limit is 2,000; the objective tolerance remains 10^{-13}. All returned solutions are non-apex. The three separate apex checks are exact self-duality tests, not additional SLSQP solves. The radius block uses 15 log-spaced distances on each sheet. The intensity block spans 10^{-3} to 10^{1}; the metric block spans $\kappa_2(W) = 1$ to 10^4 with a fixed orthogonal eigenbasis. Derivative envelopes are log-spaced from 10^{-8} to 10^{-1}.

S6. AUXILIARY GEOMETRY AND DIRECT-COORDINATE DYNAMICS

For raw polarization coordinates $w \in \mathbb{R}^2$, $r = \|w\|_2$, consider

$$\begin{aligned} F(w) &= \frac{2w}{1 + r^2}, \qquad G(w) = \frac{w}{\sqrt{1 + r^2}}, \\ P(w) &= \frac{w}{\max\{1, r\}}. \end{aligned} \tag{S20}$$

F reaches the closed disk and folds at $r = 1$. G is a smooth bijection onto the open disk. P is Euclidean disk projection and is nonsmooth at $r = 1$. Figure S3 distinguishes a finite fold from asymptotic boundary approach and shows how duplicated physical outputs can have different local sensitivity.

A. EXACT COMPARISON AT MATCHED POLARIZATION

For a radial map, use an orthonormal raw basis consisting of w/r and its perpendicular. Write $\kappa_2(A) = \sigma_{\max}(A)/\sigma_{\min}(A)$ for the spectral condition number. Direct differentiation gives Table S1.

Proposition S3 (Matched-DoLP spectra)**.** *For $0 < d < 1$, let $s_d = \sqrt{1 - d^2}$. A fixed output de has two F preimages,*

$$r_- = \frac{d}{1 + s_d}, \qquad r_+ = \frac{1 + s_d}{d} = r_-^{-1}. \tag{S21}$$

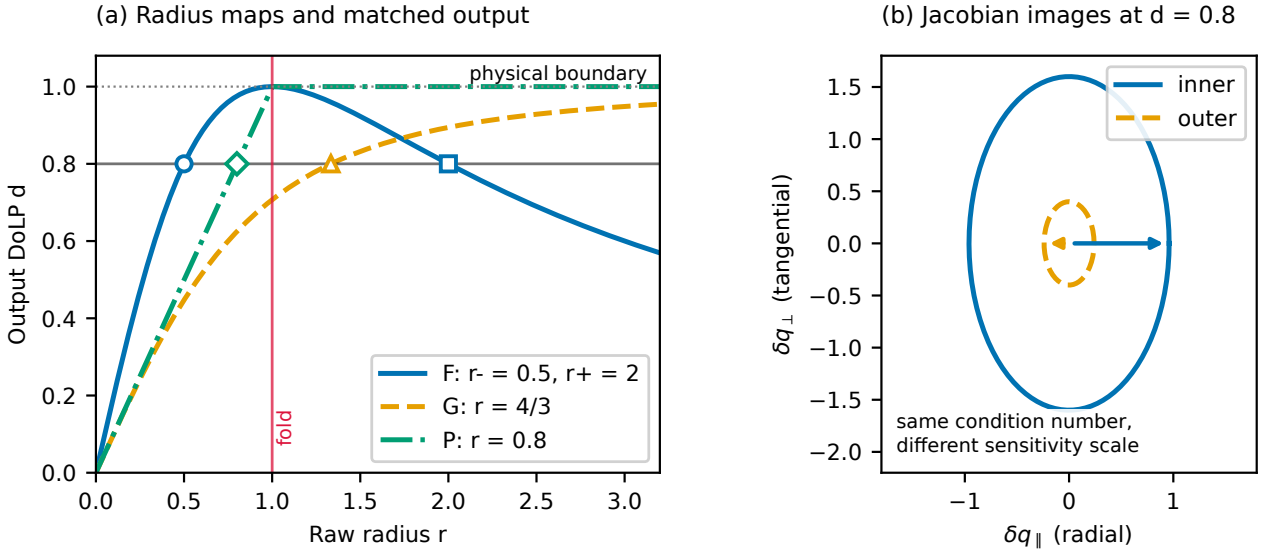


FIGURE S3. **Matched outputs and different raw-coordinate geometry. (a) Radial output degree for the folded map *F*, monotone map *G*, and disk projection *P*. At $d = 0.8$, the raw radii are 0.5 and 2 for *F*, 4/3 for *G*, and 0.8 for *P*, matching the descent diagnostic's initialization. (b) The differential images of a unit raw perturbation ball at the two *F* preimages are concentric ellipses in physical polarization-increment coordinates. Their aspect ratios agree, but the outer-sheet singular values are one quarter of the inner-sheet values. Arrows show the opposite signs of the radial eigenvalues. These are local Jacobian images, rather than finite perturbation regions.**

TABLE S1. **Exact raw-coordinate Jacobian eigenvalues and image. Here $r > 0$; the origin follows by continuity. The condition numbers below concern the normalized-polarization block at fixed intensity.**

Map	Image	Radial eigenvalue	Tangential eigenvalue
F	Closed disk	$2(1-r^2)/(1+r^2)^2$	$2/(1+r^2)$
G	Open disk	$(1+r^2)^{-3/2}$	$(1+r^2)^{-1/2}$
$P, r<1$	Disk interior	1	1
$P, r>1$	Boundary circle	0	$1/r$

Their signed radial/tangential eigenpairs are

$$\begin{aligned} F_- &: \big(s_d(1+s_d),\, 1+s_d\big), \\ F_+ &: \big(-s_d(1-s_d),\, 1-s_d\big), \\ G &: (s_d^3,\, s_d). \end{aligned} \tag{S22}$$

Consequently,

$$\kappa_2(DF) = \frac{1}{\sqrt{1-d^2}}, \qquad \kappa_2(DG) = \frac{1}{1-d^2}. \tag{S23}$$

Proof. Solve $d = 2r/(1+r^2)$ for $r_\pm$, and $d = r/\sqrt{1+r^2}$ for G. Substitution into Table S1 gives (S22). Taking singular-value ratios proves (S23). □

Thus the folded map attains the nondegenerate boundary exponent in Theorem S4. Removing its finite fold with G avoids finite boundary critical points but worsens this raw-coordinate condition number near $d = 1$.

The two F sheets have equal condition numbers, yet their singular-value scale ratio is $(1-s_d)/(1+s_d)$. For $q = H(w)$ and $J = DH(w)$, direct raw-coordinate Euclidean gradient flow gives $\dot{q} = -JJ^{\mathsf{T}}\nabla_q \ell$. The radial mobilities, the squared radial Jacobian eigenvalues, are

$$\begin{aligned} m_- &= (1-d^2)(1+s_d)^2, \\ m_+ &= (1-d^2)(1-s_d)^2, \qquad m_G = (1-d^2)^3. \end{aligned} \tag{S24}$$

As $d \downarrow 0$, $m_- \to 4$ while $m_+ \sim d^4/4$. Their ratio is asymptotic to $d^4/16$. Low-polarization outputs can therefore be nearly isotropically conditioned on both sheets while having very different absolute optimization scales. A backbone or optimizer can precondition these coordinates; these formulas are not network condition numbers.

B. A FINITE-BOUNDARY CONDITIONING BOUND

At a smooth physical boundary, a finite smooth parametrization loses a normal derivative. For the disk this follows immediately: if $H(w_0) = e$, $\|e\| = 1$, then $e^{\mathsf{T}}H(w) \le 1$ has an interior maximum at w_0, so $DH(w_0)^{\mathsf{T}}e = 0$. This is a supporting-normal form of the first-order obstruction in [3]. We now quantify the conditioning for radial heads.

Theorem S4 (Finite-boundary rate)**.** *Let $H(w) = \phi(r)w/r$. Suppose $R > 0$ has a two-sided open neighborhood on which*

ϕ is C^2, $0 \le \phi \le 1$, $\phi(R) = 1$, and $|\phi''| \le L_\phi$ for some scalar $L_\phi > 0$. For $r \to R$ with $d = \phi(r) < 1$,

$$\kappa_2(DH(w)) \ge \frac{\phi(r)}{r\sqrt{2L_\phi(1-d)}}. \tag{S25}$$

A singular Jacobian has infinite condition number. If $\phi''(R) = -c < 0$, then

$$\kappa_2(DH(w)) \sim \frac{1}{R\sqrt{2c(1-d)}}. \tag{S26}$$

Proof. Put $h = 1 - \phi$. Fermat's rule gives $h'(R) = 0$. In a sufficiently small neighborhood, $z = r - h'(r)/L_\phi$ remains in the original neighborhood: $|h'(r)| \le L_\phi |r - R|$. Taylor's upper bound and $h(z) \ge 0$ give

$$0 \le h(z) \le h(r) - |h'(r)|^2/(2L_\phi).$$

This local self-bounding smoothness argument is related to standard nonnegative-loss estimates [4]. Near R, the singular values satisfy $|\phi'(r)| \le \phi(r)/r$, since they tend to 0 and $1/R > 0$, respectively. Their ratio gives (S25) at regular points; singular points have infinite condition number. If $\phi''(R) = -c$, Taylor expansion gives $1 - d \sim c(r-R)^2/2$ and $|\phi'(r)| \sim c|r - R|$, proving (S26). □

The two-sided neighborhood is essential: the map $\phi(r) = r/R$ defined only for $r \le R$ is a counterexample without it. The exponent also requires bounded second derivative. A C^1 local profile $1 - c|r - R|^p$ with $1 < p < 2$ has a smaller exponent. The theorem identifies an attainable rate for this regularity class; the rational map is one example, not a uniquely optimal head.

C. AN EXACT STATIONARY EXAMPLE

For ideal four-analyzer measurements, $W = \mathrm{diag}(1, 1/2, 1/2)$. Let $s^* = I^*(1, de)$ with $I^* > 0$ and $0 < d < 1$. The aligned folded stationary output has

$$\begin{aligned} I_b &= I^*(2+d)/3, \qquad s_b = I_b(1, e), \\ L(s_b) - L(s^*) &= (I^*)^2(1-d)^2/6, \\ \gamma_r &= -(I^*)^2(2+d)(1-d)/9. \end{aligned} \tag{S27}$$

Its gradient is $I^*(1-d)(-1, e)/3$. The negative gradient points into the cone, yet the head's first derivative annihilates it. Supplementary Section S1 gives the complete ideal-metric critical-point classification, including the anti-aligned saddle and the monotone and disk-projection controls. Convex post-hoc projection remains a separate baseline [2]; its standard error guarantee is recalled in Supplementary Section S2.

D. DERIVATIVE CHECKS AND MATCHED INITIALIZATION

The numerical study uses NumPy 2.3.5, SciPy 1.18.1, and Matplotlib 3.11.2 in float64; it has no random optimization seed. Central differences with step 10^{-5} check 36 Jacobians at matched output degrees $0.05, 0.3, 0.8, 0.99$ and directions $0, 0.71, 2.1$ radians, plus eight full-head fold Hessians. The maximum Jacobian error is 2.00×10^{-10}, singular-value discrepancy 6.66×10^{-16}, stationary-gradient magnitude 1.67×10^{-16}, and Hessian error 7.42×10^{-11}. These checks supplement the proofs.

Figure S4(c) uses independent raw polarization coordinates and fixed intensity. The folded inner and outer initial radii are 0.5 and 2, the monotone radius is $4/3$, and the disk radius is 0.8. Their physical initial prediction is identical. Plain gradient descent uses step 0.1 for 4,000 updates of $\frac{1}{4}\|H(w) - (0.3, 0)\|^2$. The outer folded arm ends at loss 1.124×10^{-3}; the other three end below 10^{-20}. The prescribed raw coordinates and their Jacobians determine these untuned trajectories.

S7. RETAINED SENSITIVITY TO INTENSITY AND METRIC CONDITIONING

The 82-case sensitivity study supplies controlled perturbations and their error envelopes. Figure S5 shows its finite-radius error envelope, derivative-error perturbations, low-intensity amplification, and metric-conditioning sweep. The denominator in panel (b) is an independently computed gap estimate. Sections S8–S9 report a separate primal-dual numerical width for that estimate.

S8. FIRST-ORDER INVERSION AND NUMERICAL GAP BRACKETS

A. DERIVATIVE-INFORMATION REGIMES

For $w = re$, $r > 0$, the folded Jacobian has radial eigenvalue $a_r = 2(1 - r^2)/(1 + r^2)^2$ and tangential eigenvalue $t_r = 2/(1+r^2)$. It is singular only at $r = 1$. Off the fold, $j_r = IJ^\mathsf{T} g_p$ gives the main paper's first-order inverse. If the first-derivative errors are $\delta b, \delta j$, put

$$A_\phi = \begin{pmatrix} -\phi e^\mathsf{T} \\ \mathrm{Id}_2 \end{pmatrix}, \qquad T_r = A_\phi J^{-1}/I. \tag{S28}$$

The exact recovery error and an induced-norm envelope are

$$\begin{aligned} \delta g^{(1)} &= e_0 \delta b + T_r \delta j, \\ \|\delta g^{(1)}\|_M &\le k_b \varepsilon_b + \sqrt{\lambda_{\max}(T_r^\mathsf{T} M T_r)}\, \varepsilon_j. \end{aligned} \tag{S29}$$

A simpler bound follows from $\|J^{-1}\|_2 = 1/\min(|a_r|, t_r)$ and $\|A_\phi\|_2 = \sqrt{1 + \phi^2}$. The prescribed-envelope comparison uses (S29), the finite-radius naive-reuse envelope, and the model-corrected envelope described in Section S9, on the same prescribed inner radii and derivative perturbations. The two derivative orders have equal numeric envelope magnitudes for illustration; this does not impose equivalent acquisition cost or error statistics.

B. PRIMAL AND DUAL NUMERICAL CANDIDATES

For $\mathcal{Q}(u) = \frac{1}{2}\|u - z\|_W^2$, the Lorentz cone is self-dual. For every $q_\mathrm{d} \in \mathcal{C}$ and feasible u, $q_\mathrm{d}^\mathsf{T} u \ge 0$, and therefore

$$\begin{aligned} \mathcal{Q}(u) &\ge \mathcal{Q}(u) - q_\mathrm{d}^\mathsf{T} u \\ &\ge \inf_x \{\mathcal{Q}(x) - q_\mathrm{d}^\mathsf{T} x\} \\ &= -q_\mathrm{d}^\mathsf{T} z - \tfrac{1}{2} q_\mathrm{d}^\mathsf{T} W^{-1} q_\mathrm{d}. \end{aligned} \tag{S30}$$

This is the standard weak-duality construction [5], specialized to the fixed quadratic. Denote its dual value by $\mathcal{D}(q_\mathrm{d})$. For a

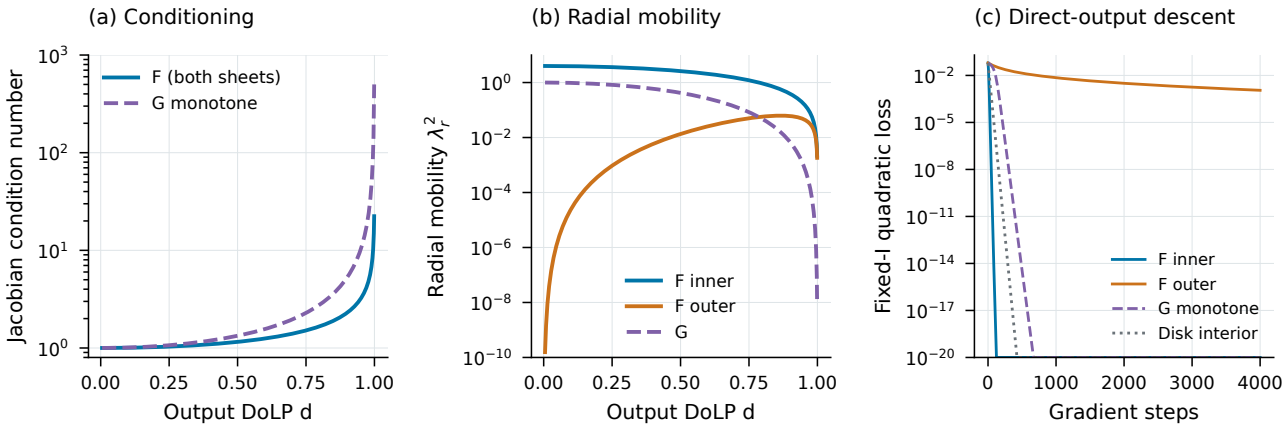


FIGURE S4. **Auxiliary conditioning and direct-output dynamics. (a) Exact normalized-polarization Jacobian condition numbers at matched output DoLP. (b) Radial mobilities: the two folded sheets have equal condition numbers and different absolute scales. (c) Fixed-intensity descent to $(I, Q, U) = (1, 0.3, 0)$, with initial normalized output $(0.8, 0)$ in every arm, quadratic polarization weight $1/2$, step 0.1, and 4,000 gradient updates. Values below 10^{-20} are clipped for display. The trajectories use the prescribed raw coordinates and a common step size.**

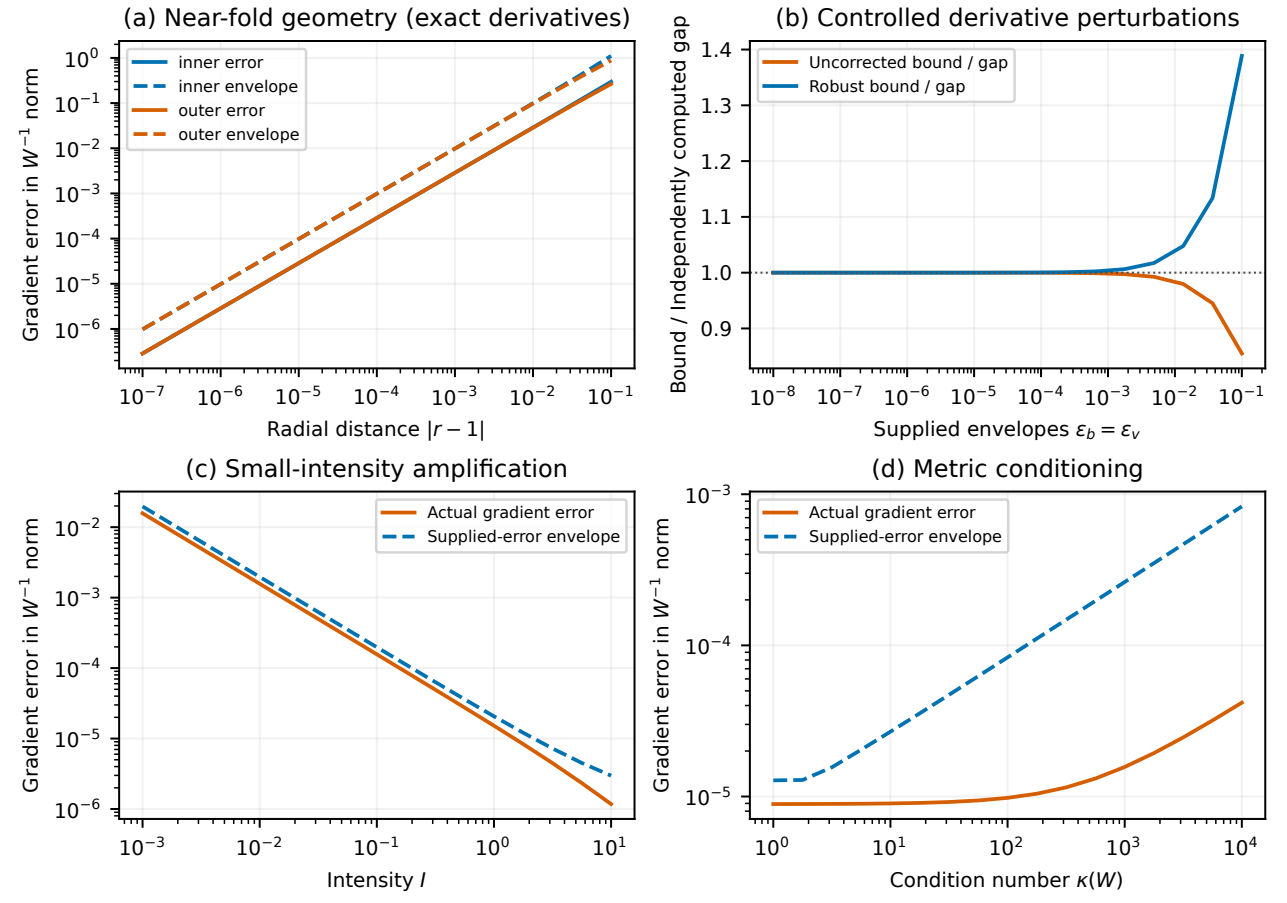


FIGURE S5. **Retained controlled sensitivity. The scalar envelopes are supplied before evaluating recovered errors. Panel (b) compares bounds with an independently computed objective-gap estimate. The metric norm changes with W in panel (d).**

feasible primal point $\bar{u}$, the objective gap lies between $\mathcal{Q}(s) - \mathcal{Q}(\bar{u})$ and $\mathcal{Q}(s) - \mathcal{D}(q_{\mathrm{d}})$; the interval width is the primal-dual objective gap.

For each saved SLSQP candidate $\widehat{u}$, the audit forms $\bar{u} =$

$\Pi_{\mathcal{C}}(\widehat{u})$ using Euclidean Lorentz-cone projection,

$$\Pi_{\mathcal{C}}(t,p) = \begin{cases} (t,p), & \|p\| \le t, \\ 0, & \|p\| \le -t, \\ \frac{1}{2}(t+\|p\|)(1, p/\|p\|), & \text{otherwise.} \end{cases} \quad \text{(S31)}$$

This is a feasibility repair for the numerical audit, distinct from the physical-metric estimate. It has maximum norm 6.80×10^{-14} across the 274 saved cases. For the dual, the code selects the largest objective among $q_{\mathrm{d}} = 0$, the cone projection of $W(\bar{u}-z)$, and a cone-feasible supporting-normal ray candidate. For $\bar{u}_p \neq 0$, the latter is $q_{\mathrm{d}} = \gamma d_{\mathrm{ray}}$, with direction $d_{\mathrm{ray}} = (1, -\bar{u}_p/\|\bar{u}_p\|)$ and coefficient

$$\gamma = \max\{0, -d_{\mathrm{ray}}^{\mathsf{T}} z/(d_{\mathrm{ray}}^{\mathsf{T}} W^{-1} d_{\mathrm{ray}})\}. \quad \text{(S32)}$$

When $\bar{u}_p = 0$, the ray candidate is omitted and the other two candidates remain; the apex optimality test uses the full self-dual cone as in Section S5. Cone feasibility of the primal and dual candidates and every recorded width are checked. Small negative slacks caused by floating-point rounding are repaired by raising the intensity coordinate to at least the polarization norm, with the numerical guard recorded in the code. Float64 widths are tested with tolerance 2×10^{-12}; all signed values are retained. These float64 results support numerical consistency; outward rounding and interval error bounds would be needed for rigorous computed enclosures.

S9. PURPOSE-SPECIFIC NUMERICAL CHECKS

A. CERTIFICATE CONDITIONS AND TIGHTNESS

Four synthetic analyzer configurations induce one ideal and three non-diagonal SPD metrics, each normalized to $\mathrm{tr}\, W = 2$. The nonideal configurations vary analyzer angle, gain, efficiency, and observation variance; their exact matrices are retained. At $I = 1.2$, three directions $0, 0.7, 1.9$ radians and seven values $\alpha \in \{-0.6, -0.1, 0.01, 0.1, 0.4, 1, 2\}$ give 84 fold states by setting $z = s - \alpha W^{-1} n$. This construction tests the certificate's conditions directly. We use $I = \exp(a)$ for this block, a regular positive coordinate covered by Theorem S5.

An independent Stokes-coordinate SLSQP solve, with analytic first derivatives, cone constraint $I - \sqrt{Q^2+U^2} \ge 0$, tolerance 10^{-13}, and limit 1,000 iterations, computes an approximation to each constrained minimum. All 84 solves succeed. Maximum cone violation is 7.88×10^{-14} and KKT stationarity residual is 2.18×10^{-7}. Raw stationarity error is 9.77×10^{-16}. Central differences of the raw gradient along the radial direction, step 10^{-4}, give curvature error 2.13×10^{-8}.

Theorem S5 classifies the 24 cases with $\alpha \le 0$ as constrained optima; their computed gaps have magnitude at most 4.73×10^{-14}. All 60 positive-α cases have negative radial curvature. Of those, 56 have feasible z and attain the certificate within 4.80×10^{-14}; the four infeasible-z cases have a strict upper bound. Figure S6 shows these checks. These finite-case tests supplement the proof.

For the nonstationary extension, keep the same four metrics, three directions, and $I = 1.2$, and use nine fixed estimates z: three interior, three boundary, and three unphysical vectors. All 108 raw gradients are nonzero. A central difference of $\nabla_w \widetilde{L}$ along e, step 10^{-4}, measures the vector HVP independently. The maximum error in $v = -Ig_p$ is 1.36×10^{-8}, and complete gradient recovery has error 1.13×10^{-8}. The independent cone solves all succeed with maximum KKT residual 1.37×10^{-7}. Using the finite-difference HVP, the largest numerical excess of the solver-computed gap over the main paper's residual bound is 1.01×10^{-9}; using exact derivatives it is 2.22×10^{-16}. The main paper's inflated bound applies when a deterministic derivative-error envelope is supplied. These 108 cases test gradient recovery away from raw stationarity, while still exactly on the fold.

B. CONTROLLED NUMERICAL SENSITIVITY

The retained sensitivity record tests the error terms with 82 deterministic cases: 30 radii $r = 1 \pm \delta$, $\delta \in [10^{-7}, 10^{-1}]$; one exact-fold baseline; 17 derivative envelopes; 17 intensities; and 17 metric condition numbers. Each case has a fixed quadratic. The supplied perturbations are $\delta b = -0.75\varepsilon_b$ and $\delta v = 0.8\varepsilon_v e$, chosen before evaluating the recovery error. The original radius study uses exact derivatives and Section S4's scalar naive-reuse envelope; the other panels use the exact fold. A conservative model-derived G_p is supplied independently of the observed recovery error. The formulas and all metric matrices are retained.

All 82 independent SLSQP solves succeed, with maximum KKT residual 1.07×10^{-7} and cone violation 9.43×10^{-14}. Central differences of the raw gradient, step 2×10^{-5}, give maximum HVP error 1.14×10^{-8}, or 2.14×10^{-10} after division by $1 + \|v\|_\infty$. Every recovered-gradient error lies within its supplied envelope; every robust bound exceeds the independently computed gap estimate. The primal-dual audit in Section S8 gives its numerical uncertainty. In all 17 deliberately directed derivative-perturbation cases, the uninflated bound falls below the computed gap estimate. These are directed perturbations of the derivative observations.

Separate checks use the analytic ideal-cone projection in 28 cases, verify the support-slack counterexample, and test the apex condition in three cases. The affine-perturbation family in the main paper's first-order ambiguity equation is checked at $\tau = -1, 0, 1$; its indistinguishable values and first raw derivatives are resolved by scalar-curvature recovery within 2.23×10^{-16}. Section S5 defines the solver residuals and the constructed derivative interface.

C. RETAINED FIRST-ORDER AND NAIVE-REUSE COMPARISON

We reuse the 30 inner/outer radius locations, now prescribing equal numerical envelopes $\varepsilon_b = \varepsilon_j = \varepsilon_v = 10^{-6}$ for the two recovery routes. The imposed errors are $\delta b = -0.75\varepsilon_b$, $\delta j = 0.8\varepsilon_j e$, and $\delta v = 0.8\varepsilon_v e$. All 30 errors lie within their analytic envelopes. With exact first derivatives, inverse recovery has maximum weighted error 1.05×10^{-9} at these radii. The retained radius records include both sheets; the main paper now plots observed AD errors separately.

D. KNOWN-METRIC SECOND-ORDER CORRECTION

The main paper subtracts $I^2 a_r J W_{pp} e$ before inverting K. The triangle inequality applies to all errors in the prescribed norm balls. We test the same 30 radius points and absolute errors as the first/naive comparison. The fixed model quantities r, e, I, I', W are treated as exact; uncertainty in them is not included in the displayed envelope. Independent 80-digit decimal central differences of the raw polarization gradient, radial step 10^{-18}, agree with the analytic HVP to 4.75×10^{-16} in Euclidean norm after float64 conversion. The exact-gradient recovery discrepancies and the prescribed-error envelopes are retained in corrected_recovery_checks.json. The correction uses the retained derivative records.

E. NONQUADRATIC OBJECTIVES AND INDEPENDENT DERIVATIVE DIFFERENCES

For the test objectives, let $s = (I, Q, U)$, $\chi = Q + 0.4U$, $c_2 = (0.3, -0.2, 0.5)$, $c_3 = (-0.4, 0.7, 0.2)$, and $\rho = \|s\|_2^2$. The fixed coefficients specify three C^∞ functions with different physical Hessians:

$$\begin{aligned} f_1(s) &= 0.2I^4 + 0.1\chi^4 + 0.3IQU + 0.05U^3 \\ &\quad + 0.4I + 0.2Q - 0.3U, \\ f_2(s) &= \exp(c_2^\mathsf{T} s) + 0.03\rho^2, \\ f_3(s) &= \log(1 + \exp(c_3^\mathsf{T} s)) \\ &\quad + 0.07IQU + 0.05\rho. \end{aligned} \tag{S33}$$

Their physical gradients are differentiated analytically in the script and independently checked by 80-digit decimal central differences, step 10^{-18}; the largest discrepancy is 4.00×10^{-36} in infinity norm. The intensities are $0.2, 1.2, 5$ and directions $0, 0.7, 1.9$ radians. Decimal direction vectors are normalized at working precision to put each evaluation on the exact fold to that precision.

For each of the 27 cases, a central second difference of the composite objective along $w = (1 + t)e$, step 10^{-12}, estimates c_r. Composite first derivatives with that scalar give maximum gradient-recovery discrepancy 1.24×10^{-23} in infinity norm. Separately, a central difference of the composite polarization gradient, radial step 10^{-18}, estimates the vector HVP; its recovery discrepancy is at most 6.75×10^{-35}. The first derivative observation matrix has numerical rank two and augmentation by the radial second-output row gives rank three in all cases. The state-dependent physical Hessians are not supplied to either fold recovery route. These examples test the general recovery identity.

For completeness, with $j = \nabla_w \widetilde{f}$ and prescribed scalar-curvature error $|\delta c_r| \le \varepsilon_c$, define

$$J_0 = \begin{pmatrix} 0_{1\times2} \\ \mathrm{Id}_2 \end{pmatrix}, \qquad q_e = (1, -e)^\mathsf{T}. \tag{S34}$$

The scalar fold formula has error $e_0\delta b + (J_0\delta j + q_e\delta c_r)/I$ and hence

$$\begin{aligned} \|\widehat{g} - g\|_M &\le k_b\varepsilon_b + \frac{\sqrt{\lambda_{\max}(J_0^\mathsf{T} M J_0)}}{I}\varepsilon_j \\ &\quad + \frac{\|q_e\|_M}{I}\varepsilon_c. \end{aligned} \tag{S35}$$

Here I, e, I', M are exact and the derivative-error bounds are prescribed.

S10. RETAINED STATIONARY CLASSIFICATION AND EXACT-FOLD ERRORS

A strict saddle means a stationary point with at least one negative Hessian eigenvalue; the Hessian may be singular.

Theorem S5 (Curvature certificate at a fold)**.** *For the fixed quadratic $L(s) = \frac{1}{2}\|s - z\|_W^2 + c$, $W \succ 0$, let (a, e), $\|e\| = 1$, be an exact stationary point of $L \circ T_F$, with $I = I(a) > 0$. Set $s = T_F(a, e) = I(1, e)$, $n = (-1, e)$, and $D_n = n^\mathsf{T} W^{-1} n$. The raw radial direction is an eigenvector of the composite Hessian. Its signed eigenvalue γ_r satisfies*

$$\nabla_s L = \alpha n, \qquad \gamma_r = -I\alpha. \tag{S36}$$

If $\gamma_r \ge 0$, s is the global constrained optimum. If $\gamma_r < 0$, the raw point is a strict saddle, and

$$0 < \Delta_{\mathcal{C}}(s) \le \frac{D_n}{2I^2}\,[\max\{-\gamma_r, 0\}]^2. \tag{S37}$$

The same upper bound is zero when $\gamma_r \ge 0$. For $\gamma_r < 0$, equality holds exactly when $z \in \mathcal{C}$.

Proof. At the fold, the full-head differential has image spanned by $(1, e)$ and $(0, e_\perp)$; its orthogonal complement is n. Raw stationarity therefore gives $\nabla_s L = \alpha n$. Along $w = (1 + t)e$, the first output derivative vanishes and the second is $(0, -Ie)$. Hence $\gamma_r = -I\alpha$. Mixed radial derivatives vanish after contraction with αn, so this is a Hessian eigenvalue, not only a directional test.

The normal cone of $\mathcal{C}$ at s is $\{\tau n : \tau \ge 0\}$. If $\alpha \le 0$, $-\nabla L$ lies in this normal cone, which is the sufficient optimality condition for the convex problem. If $\alpha > 0$, the negative gradient has a strictly inward component and s is not constrained-stationary. Finally, $s - z = \alpha W^{-1} n$ gives

$$L(s) - \min_{u\in\mathbb{R}^3} L(u) = \tfrac{1}{2}\alpha^2 D_n.$$

The constrained minimum is at least the unconstrained minimum, proving the upper bound. Since W is positive definite, the minima coincide exactly when $z \in \mathcal{C}$. □

For $\varepsilon \ge 0$ and exact stationary folds with $\gamma_r \ge -\varepsilon$, the gap is at most $D_n\varepsilon^2/(2I^2)$. Under ideal analyzers $D_n = 3$. The general implication from second-order stationarity to constrained stationarity belongs to smooth-lift theory [3]. The expression above makes the coefficient and equality condition explicit for calibrated linear-Stokes data.

A. ERROR-AWARE CERTIFICATE AT AN EXACT FOLD

Put $M = W^{-1}$, $e_0 = (1,0,0)^\mathsf{T}$, and define

$$S_e = \begin{pmatrix} e^\mathsf{T} \\ -\mathrm{Id}_2 \end{pmatrix}, \quad k_b = \sqrt{M_{00}}, \quad k_v = \sqrt{\lambda_{\max}(S_e^\mathsf{T} M S_e)}. \tag{S38}$$

Proposition S6 (Derivative-error inflation)**.** *At the exact fold, let $|\widehat{b} - b| \le \varepsilon_b$ and $\|\widehat{v} - v\|_2 \le \varepsilon_v$, with I, e, I', W exact. Recover $\widehat{g}$ by the main paper's vector fold formula and compute $\widehat{\mathcal{B}}$ by the main paper's fold residual expression using $\widehat{g}$. Then*

$$\|\widehat{g} - g\|_M \le \eta, \qquad \eta = k_b\varepsilon_b + k_v\varepsilon_v/I, \tag{S39}$$

and

$$\Delta_{\mathcal{C}}(s) \le \tfrac{1}{2}\big(\sqrt{2\widehat{\mathcal{B}}} + \eta\big)^2. \tag{S40}$$

Proof. The recovery error is $e_0(\widehat{b} - b) + S_e(\widehat{v} - v)/I$. The induced norms and triangle inequality give (S39). Use the nonnegative multiplier minimizing the residual for $\widehat{g}$ in the quadratic bound for g, and apply $\|g + \lambda n\|_M \le \|\widehat{g} + \lambda n\|_M + \eta$. This proves (S40). □

The proof uses the standard Lipschitz property of distance to a convex set. The specialized content is the explicit propagation from head derivatives. If only $|\widehat{\partial_a \widetilde{L}} - \partial_a \widetilde{L}| \le \varepsilon_a$ is supplied, then $\varepsilon_b = \varepsilon_a/|I'|$. A measured finite-difference discrepancy is not itself a guaranteed error envelope. The constants expose the $1/I$ and $1/|I'|$ amplification.

S11. LOCAL-CURVATURE INPUT CHECKS

The nonquadratic objective is f_1 in (S33), with $\chi = Q + 0.4U$. Its local polarization Hessian is

$$H_{pp} = \begin{pmatrix} 1.2\chi^2 & 0.48\chi^2 + 0.3I \\ 0.48\chi^2 + 0.3I & 0.192\chi^2 + 0.3U \end{pmatrix}. \tag{S41}$$

The reporting metric is $M = \mathrm{Id}_3$, not this Hessian. Intensities and directions are $I = 1.2$ and $e = (\cos 0.7, \sin 0.7)$. The radii are $0.9, 0.99, 0.9999, 1, 1.0001, 1.01, 1.1$. Eighty-digit decimal central differences, step 10^{-18}, check H_{pp} from the physical gradient and obtain the radial HVP from the raw gradient. Geometry is normalized at working precision. The Hessian check discrepancy is at most 4.00×10^{-37}; the composite HVP identity discrepancy after float64 conversion is at most 6.87×10^{-16}.

For curvature-input error, use $r = 0.9, 1, 1.1$ and $\Delta H = \varepsilon_H B/\|B\|_2$, where

$$B = \begin{pmatrix} 0.6 & 0.3 \\ 0.3 & -0.4 \end{pmatrix}, \qquad \varepsilon_H \in \{0, 10^{-8}, 10^{-5}, 10^{-2}\}. \tag{S42}$$

The physical loss, radius, direction, intensity, and reporting norm remain fixed. The curvature-only difference is compared with the analytic linear term. Separately, the combined-error check adds $\delta b = -0.75 \times 10^{-6}$ and $\delta v = 0.8 \times 10^{-6} e$ to the same curvature perturbation. Each combined bound uses supplied $\varepsilon_b = \varepsilon_v = 10^{-6}$ and $\varepsilon_h = \varepsilon_H$; no independence assumption is used. The maximum curvature-identity discrepancy is 4.23×10^{-16}, and all combined bounds hold with the recorded 2×10^{-13} float64 guard. These checks evaluate the identities and error coefficients in finite precision. The records are in local_curvature_checks.json; the script is local_curvature_checks.py.

S12. DIRECTIONAL CURVATURE-INPUT CHECKS

The check uses $I = 1.2$, $e = (0.6, 0.8)$, $e_\perp = (-0.8, 0.6)$, and fixed observations $b = 0.31$, $v = (0.7, -0.2)$, $h_p = (0.4, 0.6)$. Radii are $r = 1$ and $r = 1 \pm 10^{-k}$, $k = 1, \ldots, 8$. At each radius, perturb the supplied action by $0.1e$ or $0.1e_\perp$, giving 34 directional checks while keeping b, v fixed.

Eighty-digit decimal arithmetic builds J directly from the Cartesian derivatives of F and K from their radial derivative, then inverts the 2×2 matrix by its determinant. The before/after recovery difference is compared with the closed coefficients in the directional-sensitivity corollary. The largest component discrepancy is 1.20×10^{-80}; both fold changes are exactly zero in this arithmetic. The corresponding full-gradient gains are

$$\frac{\|\delta g\|_2}{\|\delta h_p\|_2} = \begin{cases} |\beta_\parallel|\sqrt{1+\phi^2}, & \delta h_p \parallel e, \\ |\beta_\perp|, & \delta h_p \parallel e_\perp. \end{cases} \tag{S43}$$

The gain discrepancy is below 1.26×10^{-79}. Component and gain checks use a 10^{-65} decimal tolerance.

A log–log fit of the measured polarization gain over $k = 4, \ldots, 8$ gives radial exponents 2.00000000109 on both sheets, and tangential exponents 1.00001367279 (inner) and 0.99998632787 (outer). Figure 2(a) in the main paper plots the closed-form gains; these directional records check the same coefficients. The executable check and full decimal coefficient records are in anisotropic_curvature_checks.py and anisotropic_curvature_checks.json.

S13. MIXED SCALAR OBSERVATIONS AND INPUT-ERROR CHECKS

The mixed check uses the three objectives defined in Section S9, intensities $0.2, 1.2$, direction $e = (0.6, 0.8)$, and radii $0.8, 0.99, 1, 1.01, 1.2$, giving 30 recovery cases. The scalar c_r is obtained by an 80-digit decimal central second difference of the raw objective, radial step 10^{-12}. The physical curvature h_{rr} is obtained by a central difference of the physical gradient along $(0, e)$, step 10^{-18}, then contracted with e. Recovery agrees with the analytic physical gradient within 1.04×10^{-24} in maximum component norm.

For each case, perturb b by -0.75ε, j by $0.8\varepsilon e_\perp$, c_r by 0.6ε, and h_{rr} by $-0.5\varepsilon_h$, where $\varepsilon = 10^{-6}$ and $\varepsilon_h = 10^{-3}$. The 30 before/after changes agree with the exact mixed error identity within 1.01×10^{-79}. The Euclidean norm envelope holds with the recorded 10^{-18} allowance for the differenced baseline.

The ambiguity check fixes $I = 1.2$, $g = (0.4, 0.2, -0.3)$, $H_{pp} = \left(\begin{smallmatrix} 0.5 & 0.1 \\ 0.1 & 0.8 \end{smallmatrix}\right)$, radii $0.8, 0.99, 1.01, 1.2$, and $\tau = \pm 0.1$. It adds the centered affine/quadratic perturbation in Proposition 2 of the main paper. All eight pairs preserve $b, P_\perp j, c_r$ to 10^{-80} while changing g; the complete first derivative changes by

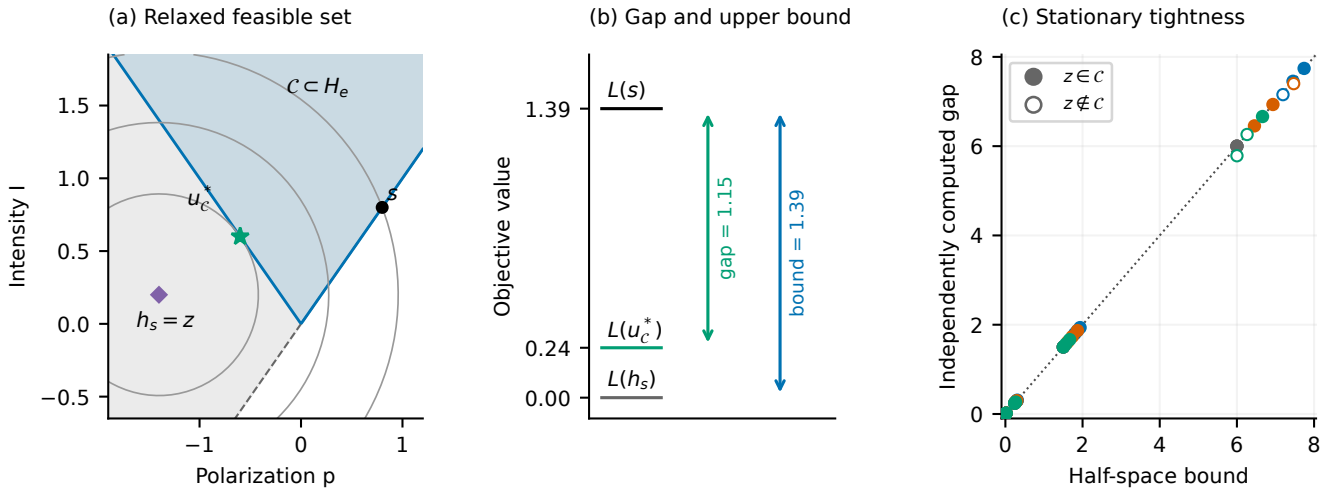


FIGURE S6. From recovered gradients to a quadratic gap bound. (a) In the (p, I) section, $\mathcal{C} \subset H_e$; relaxing the cone moves the minimum from $u_{\mathcal{C}}^* = (-0.6, 0.6)$ to $h_s = z = (-1.4, 0.2)$. The illustrative metric is diag$(1/2, 1)$ and $s = (0.8, 0.8)$. (b) Objective heights make the two differences explicit: the true gap is $1.39 - 0.24 = 1.15$, while the half-space bound is $1.39 - 0 = 1.39$. Equality holds exactly when h_s is cone-feasible. (c) The 60 negative-curvature stationary cases, with independently computed gap estimates. Filled markers have feasible z; hollow markers have infeasible z. The numerical primal-dual widths are reported separately from the analytic equality statement.

$Ia_r\tau e$. These pairs test the specified observation restriction, with physical curvature allowed to change. The implementation and inputs are in mixed_recovery_checks.py and its JSON report.

S14. COMMON-GRAPH AD ACQUISITION AND ACCURACY

The microbenchmark uses CPU eager PyTorch 2.14.1, Python 3.12.14, NumPy 2.3.5, and one intra/inter-op thread on an Apple M4 Max. Sections S17–S19 use separate continuation and repair environments. All four routes use the same quartic objective and raw head $I = \exp(a)$, $F(w) = 2w/(1 + \|w\|^2)$. Requested inputs $I_{\mathrm{req}}, r_{\mathrm{req}}$ initialize $a = \log I_{\mathrm{req}}$ and $w = r_{\mathrm{req}}(0.6, 0.8)$; geometry for recovery is evaluated from the represented coordinates.

The direct route differentiates the objective with respect to the intermediate physical Stokes tensor. The first route acquires the head gradient. The vector and mixed routes acquire head derivatives and the physical Hessian action by reverse-over-reverse AD. The mixed implementation contracts the vector HVPs to form c_r and h_{rr}. The physical-curvature query is omitted at the exact fold. Required curvature acquisition, graph construction, tensor allocation, and recovery are included in elapsed time; importing the framework and reference calculations are excluded.

Accuracy uses two precisions, intensities $0.02, 1.2$, and nine radii: 1 and 1 ± 10^{-k} for $k = 1, 3, 5, 7$. There are 144 route evaluations, including four singular first-inverse cases at the fold. The other 140 return finite gradients. The reference is the analytic physical gradient evaluated in 80-digit decimal arithmetic at the represented Stokes state, so state-rounding is accounted for in the reference. The measured errors also include finite-precision geometry arithmetic within the implementation.

Timing uses $I_{\mathrm{req}} = 1.2$, radii $0.999, 1, 1.001$, and both precisions. Each route has five warmup calls and seven batches of 20 fresh-graph calls. Method order rotates between batches. The 22 valid timing groups retain every batch mean, median, and interquartile range. At float64 and $r = 0.999$, median [interquartile] microseconds are: direct 62.9 $[62.5, 68.2]$, first inverse 114.4 $[113.4, 116.9]$, vector HVP 274.5 $[264.5, 279.7]$, and mixed 277.6 $[273.5, 284.5]$. These are observations of this acquisition implementation. The saved hardware, inputs, gradients, and batch times are in ad_route_benchmark.json. Running ad_route_benchmark.py --check-only verifies the deterministic accuracy records without repeating wall-clock measurements.

S15. SHARED-NETWORK FITTING AND DIAGNOSTIC PROTOCOL

The four analyzer rows are $A_k = \frac{1}{2}(1, \cos 2\theta_k, \sin 2\theta_k)$ for $\theta_k = 0, \pi/4, \pi/2, 3\pi/4$, so $A^{\mathsf{T}}A = \mathrm{diag}(1, 1/2, 1/2)$. Each input pixel is $y = As^* + \epsilon$, with independent Gaussian components of standard deviation 0.01. We fit the mean of $\frac{1}{2}\|As - y\|_2^2$ over all 1,024 pixels. This is a single-field fit without a held-out split.

The synthetic field uses grid coordinates $x_g, y_g \in [-1, 1]$, $I^* = 1 + 0.2\cos(\pi x_g)\cos(\pi y_g)$, $d = 0.90 + 0.09(x_g + 1)/2$, and polarization angle $\theta_{\mathrm{pol}} = \pi(0.35x_g + 0.2y_g)$. Its target is $I^*(1, d\cos 2\theta_{\mathrm{pol}}, d\sin 2\theta_{\mathrm{pol}})$. These fixed coefficients define the synthetic intensity, DoLP, and angle fields. The second field is the frozen renderer array in Section S3. The network has input dimension four, two width-32 SiLU hidden layers, and three head coordinates. Intensity is softplus plus 10^{-6}, a positive intensity floor; the two remaining coordinates use

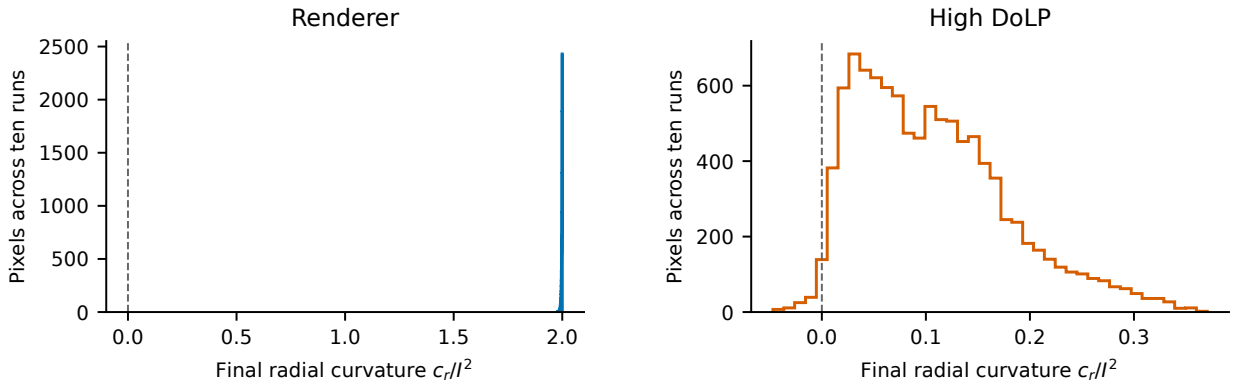


FIGURE S7. Final independent-head radial-curvature histograms across ten fixed runs per field, normalized by squared intensity. The vertical line marks zero. These are directional second derivatives at the actual trained radii, not Hessian eigenvalues of stationary folds or of the network parameter space. The near-fold incidence additionally requires the radius criterion stated in Section S15.

the rational map. The final intensity bias is inverse-softplus of one, and polarization biases are zero. The standard and broad initializations multiply the final polarization weights by 0.2 and 3, respectively. Seeds are 0–4; noise uses a separate seed offset of 1,000. Float32 CPU Adam uses learning rate 0.003, moment parameters $(0.9, 0.999)$, denominator offset 10^{-8}, zero weight decay, one thread, and 1,500 full-batch updates. All 20 runs and their final weights are retained.

Diagnostics use the per-pixel loss before averaging and float64 evaluation at stored float32 head coordinates. The radius and radial curvature are evaluated without replacing a nearby radius by one. For each of the three radius thresholds $10^{-3}, 10^{-4}, 10^{-5}$, a candidate additionally requires $c_r/I^2 < -10^{-7}$. The radius thresholds define neighborhoods; the curvature threshold excludes values within 10^{-7} of zero. We separately record the independent-head stationarity measure $\|\nabla_{a,w} L\|_2 / \max\{I^2, 10^{-12}\}$ with threshold 10^{-5}; the denominator floor prevents division by zero. Negative radial curvature at a nearby point alone does not identify a stationary saddle. Checkpoints are updates $0, 100, 500, 1000, 1500$; 25-step loss traces and all checkpoint predictions are saved. No radius/negative-curvature candidates occur at these checkpoints. Two high-DoLP pixels meet the 10^{-3} radius criterion at update 500, with nonnegative curvature. The closest final radius distances across the ten runs are 0.957049 and 0.058588 for the renderer and high-DoLP fields.

For each stored state we evaluate both signed radial offsets $0.001, 0.01, 0.05, 0.1$ at fixed intensity. Their best nonnegative loss decrease is compared with the general supporting-half-space bound including interior slack. The cone minimum uses the analytic ideal-metric solution in Section S2. No stationary-fold specialization is applied to nearby trained points. This action changes independent head outputs with the shared network fixed.

The constructed control uses $z = (1, 0.6, 0.3)$, its aligned unit direction, and stationary intensity $(2z_I + \|z_p\|)/3$. It gives $\gamma_r = -0.0976866$ and exact quadratic gap and bound 0.0180599. Its float64 Adam trajectories start either at these represented coordinates or at radial offset 0.01. Both decrease the objective over 300 updates; the nominally stationary start also escapes through finite-precision perturbations. The control checks the saddle calculation and feasible loss reduction; it is excluded from observed incidence and does not demonstrate an advantage of a deliberate perturbation over Adam.

Records are in `trained_fold_diagnostic.json` and results/trained_fold/. Each NPZ contains noisy observations, clean target, final and checkpoint outputs, raw coordinates, weights, and loss trace. Separate diagnostic NPZ files contain physical gradients, radial curvatures, stationarity measures, gaps, and bounds. Weight replay reproduces all 20 final predictions exactly. Independent AD checks of 80 head Hessian contractions have maximum discrepancy 1.78×10^{-15}. The command `trained_fold_diagnostic.py --check-only` recomputes diagnostics from saved records without retraining.

S16. BOUNDARY-DERIVED INITIALIZATION AND RESTRICTED FLOW

For a nonzero boundary Stokes state $s = I(1, e)$, any finite preimage under T_F has $F(w) = e$. The scalar relation $2r/(1 + r^2) = 1$ gives $(r - 1)^2 = 0$, hence $w = e$. If an exact preceding cone estimate has positive probability of lying on this boundary, every measurable choice of its finite raw preimage puts the same probability on $r = 1$. This conditional pushforward statement does not assume an absolutely continuous raw initialization or identify the output law of a finite solver.

For Proposition S7 below we give a self-contained statement, so this supplement can be read independently.

Proposition S7 (Restricted ideal-metric flow)**.** *Let $z_I > \rho = \|z_p\| > 0$, $W = \mathrm{diag}(1, 1/2, 1/2)$, and use any C^2 bijective intensity coordinate onto $(0, \infty)$ with $I' \neq 0$. Raw Euclidean gradient flow for $\frac{1}{2}\|T_F(a, w) - z\|_W^2$, started with $r = 1$, exists for all forward time and stays there. Except for the antipodal*

direction $e(0) = -z_p/\rho$, it converges in physical coordinates to

$$(I_b, I_b z_p/\rho), \quad I_b = (2z_I + \rho)/3. \tag{S44}$$

Its objective gap is $(z_I - \rho)^2/6$ and radial head curvature is $-I_b(z_I - \rho)/3$.

Proof. The polarization Jacobian has radial eigenvalue $a_r = 2(1 - r^2)/(1 + r^2)^2$. Therefore $dr^2/dt = -2Ia_r w^\mathsf{T} g_p$, and uniqueness of the C^1 vector field preserves $r = 1$. On this set,

$$\begin{aligned} \dot{e} &= IP_\perp z_p/2, \\ \dot{I} &= -(I')^2(3I/2 - z_I - e^\mathsf{T} z_p/2). \end{aligned} \tag{S45}$$

Write $I_- = (2z_I - \rho)/3 > 0$ and $I_+ = (2z_I + \rho)/3$. The vector field points upward below I_- and downward above I_+. Thus

$$\begin{aligned} 0 < m &= \min\{I(0), I_-\}, \\ M_0 &= \max\{I(0), I_+\} < \infty, \\ m &\le I(t) \le M_0. \end{aligned} \tag{S46}$$

The intensity coordinate a lies in the compact preimage of $[m, M_0]$; w lies on the unit circle. The vector field is smooth on a neighborhood of this compact set, so the solution extends to every forward time. In particular, $|I'|$ is bounded above and away from zero along it.

Set $u = e^\mathsf{T} z_p/\rho$. Then $\dot{u} = I\rho(1-u^2)/2$. If $-1 < u(0) < 1$,

$$\operatorname{artanh} u(t) = \operatorname{artanh} u(0) + \frac{\rho}{2}\int_0^t I(\tau)\, d\tau. \tag{S47}$$

The integral grows at least as mt, so $u(t) \to 1$. Initial $u = 1$ is already aligned. Since $u \to 1$, for any $\varepsilon > 0$ the intensity derivative eventually points toward the interval $[I_+ - \varepsilon, I_+ + \varepsilon]$ and has magnitude bounded away from zero outside it on the compact intensity range. Scalar comparison yields $I \to I_+$. The case $u(0) = -1$ stays antipodal and instead gives $I \to I_-$. Its gap and radial curvature are $(z_I + \rho)^2/6$ and $-I_-(z_I + \rho)/3$.

Because z is cone-interior, its quadratic loss is the cone minimum. For the aligned limit,

$$\begin{aligned} L(I_b, I_b e) - L(z) &= \tfrac{1}{2}(I_b - z_I)^2 + \tfrac{1}{4}(I_b - \rho)^2 \\ &= \frac{(z_I - \rho)^2}{6}. \end{aligned} \tag{S48}$$

At $r = 1$, the radial second derivative is $-Ie^\mathsf{T} g_p = -I(I - \rho)/2$, yielding the displayed curvature at I_b. □

This result specializes the differential geometry of the rational ball lift [1], [3]. Theorem 2.4, Lemma 3.24, Proposition 3.26, and Theorem 3.23 of [3] describe the first-order obstructions and second-output derivatives. Here the calibrated quadratic bound, including its support slack, supplies the approximate stopping test. Section S20 extends gap control to strongly convex objectives and compares it with quantitative squared-variable transfer.

The random-initialization theorem of [6] assumes an absolutely continuous initialization, a C^2 objective with Lipschitz gradient, and a fixed step smaller than the reciprocal Lipschitz constant. Those hypotheses do not follow from a boundary-derived law, and the theorem concerns asymptotic gradient descent rather than finite BFGS or LBFGS termination. For plain discrete gradient descent, $\nabla_w L = IP_\perp g_p$ at $r = 1$ is tangent, so one step gives $r_{\text{new}}^2 = 1 + \eta^2 I^2 \|P_\perp g_p\|^2$. Coordinatewise or quasi-Newton preconditioning need not preserve that tangency.

S17. FIXED TWO-FRAME VERIFICATION PROTOCOL AND COMPLETE CONTROLS

The machine-readable protocol, its registration receipt, raw input records, solver outputs, and independent audits are in repro/warm_start/. No optimization condition was added after viewing the grid results. The 60 input rows use seeds 0–4, angles $0, 0.7, 1.3$, noise SD $0.001, 0.01$, and next DoLP $0.8, 0.99$. Each seed starts the generator at $10000 + \text{seed}$ for each condition, so its two four-dimensional noise vectors are reused. The previous clean state is $(1, \cos 2\theta, \sin 2\theta)$; the next is $(1, d\cos 2\theta, d\sin 2\theta)$. For angle changes $\delta = 0, 0.05, 0.2$, only the next clean direction changes to $\theta + \delta$, retaining the stored next-frame noise exactly.

A. SOLVERS, PRECISION, AND TERMINATION

Every previous fit starts at $(0, 0.2, -0.1)$ with exponential intensity. BFGS, unbounded L-BFGS-B, and trust-exact use analytic head gradients, maximum 500 iterations, and gradient tolerances $10^{-4}, 10^{-6}, 10^{-8}$. L-BFGS-B additionally uses ft ol=0, maximum 5,000 function calls and 40 line-search steps. The trust-region method acquires the full analytic 3×3 Hessian. It uses its native gradient norm; the quasi-Newton methods use their native stopping rules. Previous fits with identical observations, method, and tolerance are cached, giving 270 distinct previous fits for the 1,620 continuations. Previous failures would be retained; none occurs in these SciPy arms.

The SciPy run uses macOS arm64, Python 3.12.3, NumPy 2.5.4, and SciPy 1.18.1. Initialization of a separate BFGS probe with float32 coordinates returns float64 coordinates and float64 oracle gradients; incoming oracle coordinates remain float32 in that probe. The oracle explicitly evaluates in float64. This mixed computation is not used as evidence of full float32 optimization.

The separate genuine-precision control uses Python 3.11.6, NumPy 2.3.5, and PyTorch 2.9.1, CPU, one thread, with all optimizer parameters, objective values, and gradients in float32 or float64. LBFGS has learning rate 1, history size 20, maximum 500 iterations and 1,000 evaluations, gradient tolerance 10^{-6}, zero change tolerance, and strong-Wolfe line search. It exposes no native success flag. We therefore report native-gradient acceptance, nonacceptance, and objective gap separately. Its diagnostic controller, radial trial selection, and final reference gap use float64 at the represented native coordinates with the original observations. The 22 float32 previous-fit nonacceptances are row counts representing 11 of 30 distinct previous fits. They remain in the table rather than being excluded. Float64 previous fits all meet the operational acceptance test.

B. REFERENCE, DIRECT STOPPING, AND CURVATURE STOPPING

The reference uses the ideal-metric cone optimum: for unconstrained estimate z, $\rho = \|z_p\|$, return z when $z_I \geq \rho$, the apex when $2z_I + \rho \leq 0$, and otherwise $(I_c, I_c z_p/\rho)$ with $I_c = (2z_I + \rho)/3$. An independent physical-coordinate SLSQP calculation starts at $(1.1, 0.1, 0.1)$ and uses $I \geq 0$, $I^2 - \|p\|^2 \geq 0$, objective tolerance 10^{-13}, and at most 1,000 iterations. Across all 180 distinct next-frame observation vectors, all solves succeed; maximum objective discrepancy is 5.00×10^{-14} and maximum cone violation is 7.67×10^{-15}.

The direct physical stopping test minimizes the supporting-half-space bound over its nonnegative multiplier and accepts a finite bound at most 10^{-6}. We use the support slack at the actual radius, including interior outputs. Every inaccurate SciPy endpoint is rejected; no finite accurate endpoint is rejected in this grid. This does not make the half-space relaxation exact for arbitrary data. The expression is evaluated in float64; the empirical acceptance counts are not outward-rounded numerical certificates. At the 149 near-fold SciPy endpoints, mixed recovery using the known $h_{rr} = 1/2$ agrees with the direct physical gradient within 4.17×10^{-17} componentwise. No physical-gradient availability restriction is imposed.

The local curvature gate uses $|r - 1| \leq 10^{-3}$ and $c_r/I^2 < -10^{-7}$ at the represented endpoint. Both signed radial trials and the current point are evaluated; the lowest loss is restarted once with the same solver and settings. The three trial evaluations obtain both function and gradient in this implementation. For each of the 60 zero-angle-change fits at tolerance 10^{-6}, the direct and curvature controllers also evaluate the endpoint state. Gated BFGS adds 280 function/gradient pairs; a radius-first implementation requires 17 full Hessians. Unconditional restart adds 479 pairs. The corresponding L-BFGS-B counts are 320 pairs and 21 required Hessians versus 496 pairs. The trust-region arm has no trigger, five required audit Hessians, and 240 unconditional additional pairs plus 60 unconditional solver Hessians. Coverage logging instead computes a Hessian at all 60 endpoints in each of these groups, including points outside the gate. The saved accounting separates actual analysis work, optimizer calls, final checks, and queries required by a deployed gate. These counts are not interchangeable units of work.

At tolerance 10^{-4}, the gate detects 14 of 68 false successes among the 540 SciPy continuations and also triggers at one accurate accepted endpoint. After restarting, 58 false successes remain, compared with 32 under unconditional restarting. At tolerance 10^{-8} and zero angle change, BFGS reports seven native failures, including six nonfinite terminal losses; unconditional restarting leaves six failures and the gate leaves seven. All failures are retained. Nonfinite numbers are serialized as null together with failure flags, warnings, and evaluation counts; a null maximum is never interpreted as zero gap. The file audited_summary.json explicitly separates finite and nonfinite errors, and failure_records.json preserves their conditions and terminal states.

C. INDEPENDENT REPRODUCTION AND SCOPE OF THE EVIDENCE

A new Cartesian Jacobian/Hessian implementation reproduces the original BFGS false-success IDs $1, 2, 3, 21, 25, 29, 40, 41, 45, 47, 53, 55$ (zero-based), the separate failure at ID 52, and the post-trigger counts zero and one. The largest previous-coordinate discrepancy is 5.62×10^{-14} and the largest objective-gap discrepancy is 4.86×10^{-12}. Independent PyTorch AD at all 60 frozen endpoints gives maximum gradient and Hessian discrepancies 9.49×10^{-18} and 4.45×10^{-16}. Thus the reference arithmetic, derivative identities, and solver reproduction are checked separately.

These results quantify how physical or recovered derivatives audit a local head stopping decision. The direct cone solution solves this three-variable calibrated quadratic without a raw warm start, while the direct physical bound rejects failures away from the fold that the local curvature gate misses. The 20 MLP fits in Section S15 and their zero-candidate result are unchanged and are not pooled with the two-frame conditions.

S18. PHYSICAL-REFLECTION AND RELEASED-OBSERVATION PILOTS

A. FRESNEL SEQUENCE AND CONTROLS

The additional fixed physical simulation uses a single air-to-dielectric interface with real refractive index 1.5. For incidence θ, Snell's law gives $\sin\theta_t = \sin\theta/1.5$. The squared Fresnel amplitude coefficients give reflected fractions R_s, R_p. For unpolarized illumination, specular intensity and linear-polarization magnitude are proportional to $(R_s + R_p)/2$ and $(R_s - R_p)/2$. We normalize specular intensity to one at Brewster's angle and add constant unpolarized backgrounds $0, 0.02, 0.1$. These backgrounds are not geometry-dependent diffuse-shading models.

Incidence follows $\arctan(1.5) + 18^\circ \cos(2\pi t/40)$ for 41 frames. Polarization orientation is either zero or $0.05\sin(2\pi t/40)$ radians. Photon scales are 1 000 and 10 000, and seed indices are zero and one. Each configuration has its own NumPy seed sequence $[20261022, \text{configuration index}, \text{seed index}]$. Analyzer electron counts have means NA_s, independent Poisson fluctuations, and independent Gaussian read noise of standard deviation two electrons; dividing by N gives the fitted observations. No clipping, shared frame noise, or initialization from physical truth is used. Each solver/arm begins at raw coordinates $(0, 0.2, -0.1)$ and propagates its own subsequent estimate.

The 24 sequences contain 984 observations per arm. BFGS and L-BFGS-B use the options in Section S17 at tolerance 10^{-6}. Native false successes number nine and seven, with zero native failures. Curvature-audited and physically audited branches each have zero false successes and seven interventions per solver. A radius-first curvature controller requires 22 initial endpoint Hessians per solver. Optimizer function/gradient pairs, including restart trials, total 11 394 and 11 292 in the curvature-audited branches, versus 11 813 and 11 575 without intervention. These counts do not equate a

Hessian query with a first-derivative query or establish wall-clock superiority. Direct physical auditing requires no Hessian; the script's extra diagnostic logging is separately identified.

The direct-cone truth Stokes RMSE is 0.0251904, compared with 0.0252375 for raw BFGS and 0.0251904 after either audit. Mean error is dominated by measurement noise. False successes occur with background zero or 0.02, in both orientation modes; none occurs with background 0.1 in this grid. The displayed sequence is selected by the largest raw BFGS objective gap. It is an illustrative worst case, not a typical case.

A monotone-map BFGS control under the same absolute head-gradient tolerance reports 160 native false successes. The first recorded version shares the raw initial coordinate, hence not the physical initial state. A separately recorded correction uses $w_G = q/\sqrt{1-\|q\|^2}$ at the same initial physical polarization $q = F(0.2, -0.1)$; it also reports 160. Both controls and their protocols are retained. This finite-tolerance comparison involves different map conditioning, with the monotone boundary attainable only at infinite radius; it does not establish a universal ranking. Independent physical-coordinate SLSQP checks of 20 minima have maximum loss discrepancy below 2×10^{-14}. PyTorch float64 AD at all 16 raw false successes and 48 initial control endpoints has maximum normalized radial-curvature discrepancy below 7×10^{-16}.

B. RELEASED DATA, SELECTION, AND OBSERVATION UNITS

The acquisition plan fixes the data before any solver outcomes are inspected. From the DeepSfP archive linked by its data documentation, we select the first three distinct indoor test objects: `box_f`, `father_christmas_f`, and `horse_l`. The archive's test list identifies all three. Each MAT file contains four-angle images, a foreground mask, and reference surface normals; those normals are not Stokes truth and are not used. The selected archive's README and current repository documentation describe different full split totals. We therefore identify the exact archive and selected members, rather than treating the small subset as the complete current benchmark.

PolarNS contributes object/241015104249 and scene/2410 21103114, the lexicographically first records in each category. Their `img_mean`, `img_variance`, and `image_count` arrays describe 10 000 and 1 017 captures. We use their means as released observations, without synthesizing new noisy frames. Means are divided by 4 096, matching the authors' MATLAB convention. The 2×2 polarizer layout gives $90°, 45°$ on the first row and $135°, 0°$ on the second. This spatial sampling is not a calibrated co-location claim.

The PolarBurstSR subset is test/0000000 through test/0 000002, with `frame_00.png` through `frame_13.png`. The authors' real-data configuration points to this released test directory. Each raw image is 192×192 uint16; division by 65 535 and unpacking of each 4×4 sensor block follow the provided loader exactly. Four nominal color planes, R, G1, G2, and B, each yield 48×48 four-analyzer observations without demosaicing. A track uses one fixed sensor coordinate across the 14 frames. Because no registration is performed, it need not follow one scene point. The released high-resolution references are not used for aligned reconstruction metrics.

For static images, eligible pixels are finite, nonnegative, have every analyzer channel below 0.995, and have nominal intensity at least 5% of the foreground/sample 99th percentile. DeepSfP additionally requires its foreground mask and retains its stored normalized double units. A seeded uniform selection chooses 128 eligible pixels per image; the seed sequence is $[20261022, \text{sample index}]$. The enriched selection takes 64 evenly spaced ranks from the top 1% by observed DoLP, with a minimum pool of 64. Each selection is sorted in raster order and forms a separate spatial continuation stream. This is an implementation stress test on static observations, not a video trajectory.

For each burst color plane, the same finite, saturation, and relative-intensity conditions are evaluated only on the first frame. A fixed 4×4 spatial grid contributes its eligible positions. Eight positions are taken at evenly spaced ranks from the top 1% first-frame observed-DoLP pool, with a minimum pool of eight. All 14 frames at each selected position are retained. After eligibility, there are 167 uniform and 96 enriched tracks, totaling 3 682 burst entries per solver/arm. Uniform and enriched selections may overlap, and color planes and frames share a capture. They are separate conditional selections, not independent scene samples or estimates of population incidence.

C. FITTING, DIAGNOSTICS, AND COMPLETE COUNTS

The nominal ideal analyzer matrix and $W = \mathrm{diag}(1, 1/2, 1/2)$ define the same least-squares objective and exact cone reference as Section S17. The units above are fixed per data source; no per-pixel normalization or clipping is introduced. No Poisson-likelihood, real-camera calibration, denoising, super-resolution, or normal-estimation improvement is inferred from objective-gap accuracy.

Each solver uses four arms: cold initialization, uncorrected warm start, curvature-triggered restart, and physical-bound-triggered restart. A stream starts at $[\log\max(\widehat{I}, 10^{-6}), 0.2, -0.1]$, with $\widehat{I} = \sum_k y_k/2$. Cold starts recompute this point for each observation. Other arms propagate their own preceding raw state, including native failures. Solver settings, local radius/curvature thresholds, objective gap 10^{-6}, and the one-restart rule remain those in Section S17. The protocol is recorded before optimization; no additional conditions are selected after seeing the results.

All 37 136 fits are retained. The fixed curvature controller has no intervention and detects none of the raw warm-start false successes. The physically audited branch intervenes 69/64 times in the PolarNS means and 16/101 times in the bursts. It rejects all inaccurate final endpoints, although a single radial restart does not repair all of them. Cold starts reduce the counts in this subset but retain 14 L-BFGS-B false successes, so changing initialization is not a general physical stopping criterion.

Among raw warm-start false successes, PolarNS radii range from approximately 0.996 to 6.74×10^6, and burst

TABLE S2. Complete released-data selection counts. Each pair is BFGS/L-BFGS-B; *n* is the number of entries per solver/arm. "Physical post" counts inaccurate *native* successes after the physically audited restart branch. Every such final endpoint is rejected by its physical bound; it is not an accepted output of the monitor. All native failure counts are zero.

Subset and selection	n	Cold	Warm	Curvature post	Physical post
DeepSfP, uniform	384	0/0	0/0	0/0	0/0
DeepSfP, high observed DoLP	192	0/0	0/0	0/0	0/0
PolarNS means, uniform	256	0/0	43/0	43/0	0/0
PolarNS means, high observed DoLP	128	0/7	68/64	68/64	68/64
PolarBurstSR, uniform	2 338	0/1	0/9	0/9	0/9
PolarBurstSR, high observed DoLP	1 344	0/6	16/95	16/95	16/90

radii from 0.752 to 314. No inaccurate endpoint meets the fixed local fold gate. The physical bound rejects all 295 raw warm-start false successes and has zero inaccurate acceptance across the recorded arms. Independent SLSQP checks of 29 reference minima have maximum objective discrepancy 1.31×10^{-18}. Independent PyTorch float64 AD checks all 309 cold/warm false successes and 32 initial controls; maximum head-gradient, Hessian, and physical-gradient differences are 7.77×10^{-18}, 8.89×10^{-16}, and 1.39×10^{-17}. No native solver failures, warnings, or nonfinite oracle calls occur in this fixed run.

The curvature controller requires a Hessian only after its radius guard. The scripts also evaluate Hessians for full endpoint coverage and independent final diagnostics; these are verification overhead. Physical-gradient auditing requires zero Hessians. Function/gradient, restart-trial, and diagnostic queries remain separate; optimizer-only timings are not end-to-end runtime comparisons. Exact archive-member names, verified CRC32 and SHA-256 values, byte-range receipts, selected coordinates, input observations, native flags, endpoint states, bounds, query counts, protocols, and figure generators are included in the reproduction records. Raw data are obtained from the original providers; no full multi-gigabyte archive or pretrained model is required for this pilot.

S19. FROZEN COORDINATE-REPAIR FOLLOW-UP

This follow-up was designed after the released-data results in Section S18 were inspected. The code and protocol were hashed before the new algorithm outcomes were evaluated. All 4 642 released observations, 273 streams, sampling strata, nominal measurement units, and ordering are reused unchanged. All 24 Fresnel sequences and 984 frames are also reused; they are a previously inspected mechanism control, not a held-out test set. No additional seeds, thresholds, data, or parameter grids are introduced. The experiment uses float64 NumPy 2.5.4 and SciPy 1.18.1 on one local CPU process.

A. SHEET SYMMETRY AND ITS SCOPE

Let $C(w)=w/\|w\|^2$ for $w\neq0$, $r=\|w\|$, $e=w/r$, and $H_e=\mathrm{Id}_2-2ee^{\mathsf{T}}$. Direct substitution and differentiation give

$$F(C(w))=F(w),\qquad DC(w)=r^{-2}H_e. \tag{S49}$$

For $h(a,w)=f(T_F(a,w))$, at the same intensity and observations,

$$\nabla_w h(a,C(w))=r^2H_e\nabla_w h(a,w). \tag{S50}$$

Indeed $h(a,w)=h(a,C(w))$, so the chain rule gives $\nabla_w h(a,w)=r^{-2}H_e\nabla_w h(a,C(w))$, and $H_e^2=\mathrm{Id}_2$. The normalized-output condition number is unchanged, while the polarization-gradient norm increases by r^2 when an outer state is moved inward. The intensity derivative is unchanged. These identities follow from the two-sheet map and the matched spectra in Section S6. They improve derivative scale at outer coordinates while retaining the fold and intensity dependence.

B. FIXED ALGORITHM AND CONTROLS

All head solvers use gradient tolerance 10^{-6} and at most 500 iterations. L-BFGS-B additionally uses zero objective tolerance, at most 5 000 function evaluations, and 40 line-search steps. Every new solve resets its approximate inverse Hessian and line-search history. At the first released observation of each stream, and at every released cold start, the initialization is $[\log(\max\{S_{0,\mathrm{LS}},10^{-6}\}),0.2,-0.1]$. The corresponding Fresnel initialization is $[0,0.2,-0.1]$. Every continuation arm retains its own preceding result, including a rejected result.

The switch-before-solve arm replaces an outer preceding state by $(a,C(w))$ before the next solve; it otherwise uses native acceptance. The audited coordinate arm first solves from the preceding state and evaluates Corollary 5's physical bound with the exposed physical gradient. On rejection, it uses the following fixed choices. If $|r-1|\leq10^{-3}$, one full analytic head Hessian supplies c_r. If $c_r/I^2<-10^{-7}$, the candidates are the current point and radial offsets $-0.01,+0.01$; the lowest-loss candidate is moved to the inner sheet if necessary and starts one fresh solve. If the negative-curvature condition does not hold and $r>1$, the endpoint is moved to the inner sheet and starts one fresh solve. Otherwise no head correction is made. A final physical audit retains any rejection. There is at most one head restart per observation. The audited cold control instead uses one fresh cold solve on the same type of rejection.

The guarded coordinate arm applies standard projected gradient only if that coordinate repair remains rejected. For

the ideal quadratic, write $g(s) = W(s - z)$ with $W = \mathrm{diag}(1, 1/2, 1/2)$. Its update is

$$s^{k+1} = \Pi_{\mathcal{C}}\big(s^k - g(s^k)\big). \tag{S51}$$

Here $\Pi_{\mathcal{C}}(v) = \arg\min_{u\in\mathcal{C}} \|u-v\|_2^2$ is the Euclidean projection in (S31). The W-projection P_W in Section S2 instead supplies the direct least-squares reference. For example, $\Pi_{\mathcal{C}}(0,1,0) = (1/2, 1/2, 0)$, whereas $P_W(0,1,0) = (1/3, 1/3, 0)$. The projected-gradient step size is one because $\lambda_{\max}(W) = 1$. Constrained optimality gives $s^* = \Pi_{\mathcal{C}}(s^* - g(s^*))$. Euclidean nonexpansiveness therefore yields

$$\|s^{k+1}-s^*\|_2 \le \|\,\mathrm{Id}_3 - W\|_2 \|s^k-s^*\|_2 = \tfrac{1}{2}\|s^k-s^*\|_2. \tag{S52}$$

This standard contraction follows from nonexpansiveness of projection and the fixed quadratic metric. The implementation audits before and after updates, reuses the acquired physical gradient, stops only at bound at most 10^{-6}, and permits at most 64 updates. An unresolved bound is reported as a rejection, never as a repaired accepted solution. The finite budget alone is not an unconditional accuracy guarantee. In exact arithmetic, Corollary 5 implies gap at most 10^{-6} whenever this bound accepts; the recorded float64 calculation is not outward-rounded certification.

A positive-intensity physical correction is lifted by the inner inverse $w = q/(1 + \sqrt{1 - \|q\|^2})$, $q = p/I$, $a = \log I$. Only the square-root argument is clamped to remove feasibility roundoff. The observed lift discrepancy in the independent audit is at most 1.12×10^{-16}. An exact apex has no finite exponential lift: the specified implementation would retain its physical output and cold-start the next head solve. No experimental output requires this case. The standalone projected-gradient control starts from the same first physical state and subsequently carries its own physical result; it uses the same updates, bound, and budget without any head solve. The direct cone reduction is timed separately and supplies the reference gap; its answer is never used to initialize an iterative arm.

C. FULL ACCOUNTING AND PAIRED AUDIT

The follow-up stores 78 764 sequential output records: 67 512 head-arm outputs across six strategies and two solvers, and 11 252 outputs from the two physical controls. These counts exclude extra internal restarts. All native head invocations return success, with no recorded warnings or nonfinite queries. Monitored acceptance and native head status are separate fields; a physical fallback does not inherit a native head success flag.

Table S3 retains the released sampling strata. The standalone physical controls and the guarded coordinate arm have no inaccurate outputs in any stratum. In the released grid the guarded arm makes 16/25 outer restarts and two/15 physical fallbacks; its total projected updates are three/22. The audited cold control makes 18/35 cold restarts. The audited coordinate arm makes 18/33 outer restarts and leaves 7/22 inaccurate outputs, all rejected. The switch-before-solve arm performs 26/41 sheet changes and leaves 9/16 inaccurate native successes. No real-data correction requires a Hessian query. In the Fresnel grid each guarded arm makes seven fold restarts, acquires seven full Hessians and 21 trial FG pairs, and uses no projected fallback. Those full-Hessian queries are not reported as cheaper directional-HVP implementations.

The paired audit keeps all 9 284 historical raw warm endpoints as a common starting population. Its repair selection is the physical bound rejection, not the reference objective gap. All 295 rejections in this population are the previously inaccurate native successes. Audited coordinate repair leaves 7/39, using 124/157 outer restarts and 1 919/1 259 additional native FG pairs. Guarding this procedure adds seven/39 physical fallbacks and nine/56 projected updates, leaving zero inaccurate outputs. Audited cold restarting leaves 0/5, using 127/168 restarts and 3 421/1 407 FG pairs. These conditional endpoint counts differ from the sequential experiment because an intervention changes later initialization.

Independent PyTorch 2.9.1 float64 differentiation checks all 295 historical false-success endpoints and 32 evenly spaced warm controls. Their maximum output-preservation discrepancy is 1.74×10^{-18} and gradient-transformation discrepancy 7.78×10^{-16}. A separate physical-step implementation checks those same 327 states: all reach the bound threshold within nine updates, no objective increase is observed, and the maximum excess over the half-distance contraction is 5.56×10^{-17}. The separate implementations check the derivatives and physical updates numerically.

Native FG pairs, physical-gradient queries, projected updates, full Hessians, trial evaluations, coordinate changes, internal solver calls, and observed wall times are retained separately. Post-hoc reference-gap calculations are excluded from algorithm timings and query counts. Timings are one sequential CPU pass with fixed arm order; the protocol does not support statistical timing claims. The results quantify partial coordinate repair and the effectiveness of standard physical controls on these previously inspected quadratics. Evaluation on unseen scenes and aligned imaging-quality metrics would address different questions. Exact frozen-summary replay and source hashes are included in application/chart_repair/.

S20. PROOFS AND CHECKS FOR FINITE-TOLERANCE STOPPING

Write $\Delta(s) = f(s) - \min_{u\in\mathcal{C}} f(u)$ and $x_+ = \max\{x, 0\}$. The guarantee concerns a C^2, μ-strongly convex physical objective on $\mathbb{R}^3$. The evaluated Stokes state has $0 < I \le \bar{I}$ and $0 \le r \le 1$. The radial physical Hessian is bounded above by $L \ge \mu$ at the evaluated point when $r \ge 1/2$. A global Hessian upper bound is sufficient but not necessary for this pointwise assumption. The intensity coordinate may be any regular coordinate specified in the main text. All identities below use its exact evaluated I, I' and exact r, e. Uncertain geometry requires an additional error model.

TABLE S3. Inaccurate released outputs by original stratum. Pairs are BFGS/L-BFGS-B. "Switch" applies before every warm solve; "chart" means audited coordinate repair without physical fallback. Guarded coordinate repair, standalone physical projected gradient, and the direct cone control are zero in every row.

Subset and stratum	Entries	Warm	Switch	Audited cold	Audited chart
DeepSfP, uniform	384	0/0	0/0	0/0	0/0
DeepSfP, high DoLP	192	0/0	0/0	0/0	0/0
PolarNS means, uniform	256	43/0	3/0	0/0	2/0
PolarNS means, high DoLP	128	68/64	1/2	0/0	0/9
Burst, uniform	2 338	0/9	0/2	0/0	0/1
Burst, high DoLP	1 344	16/95	5/12	0/2	5/12

A. SUPPORTING-GAP INEQUALITY FROM STRONG CONVEXITY

Let $s = I(1, \phi e)$, $n = (-1, e)$ and $\sigma = I(1-\phi)$. For every $u \in \mathcal{C}$, $n^{\mathsf{T}}(u-s) \le \sigma$. Strong convexity gives, with $d = u - s$,

$$f(s) - f(u) \le -g^{\mathsf{T}}d - \frac{\mu}{2}\|d\|^2 \le -(g+\lambda n)^{\mathsf{T}}d - \frac{\mu}{2}\|d\|^2 + \lambda\sigma. \tag{S53}$$

Taking the supremum over $d \in \mathbb{R}^3$ yields $\Delta(s) \le \|g + \lambda n\|^2/(2\mu) + \lambda\sigma$ for every $\lambda \ge 0$. This is the standard strong-convexity/dual-residual argument. The original quadratic corollary is sharper when the full metric is known; this isotropic inequality extends to the present strongly convex class.

B. A POINTWISE BOUND USING ONE-SIDED RADIAL CURVATURE

For $r > 0$ write $g_p = me + u$, $u \perp e$. The first derivatives satisfy

$$b = g_I + \phi m, \qquad j/I = a_r m e + t_r u. \tag{S54}$$

Suppose $|b| \le \varepsilon$ and $\|j\|/I \le \varepsilon$. Then $|a_r m| \le \varepsilon$, $\|u\| \le \varepsilon/t_r$. For $r < 1$, setting $\lambda = 0$ in (S53) gives

$$U_1 = \frac{1}{2\mu}\big[(\varepsilon + \phi\varepsilon/a_r)^2 + (\varepsilon/a_r)^2 + (\varepsilon/t_r)^2\big], \qquad \Delta(s) \le U_1. \tag{S55}$$

At $r = 1$ define $U_1 = +\infty$. At the origin $\phi = 0$ and $DF = 2\,\mathrm{Id}_2$; the displayed first-order estimate remains a conservative bound with $a_r = t_r = 2$ and does not require a radial direction.

For $1/2 \le r \le 1$ let $k_r = |\kappa_r|$ and assume $c_r \ge -I\nu$. Since $\kappa_r < 0$, the chain rule implies

$$m \le M_r := \frac{\nu + ILa_r^2}{k_r}. \tag{S56}$$

Define the continuous slack-to-derivative ratio

$$D_r = \frac{(1-r)(1+r^2)}{2(1+r)}. \tag{S57}$$

For $r < 1$ this equals $(1-\phi)/a_r$, while $D_1 = 0$. If $m \ge 0$, choose $\lambda = 0$ and bound m by M_r. If $m < 0$, choose $\lambda = -m$; then $g + \lambda n = (b + (1-\phi)m, u)$ and $\lambda\sigma = -mI(1-\phi)$. Consequently,

$$\begin{aligned} U_+ &= \frac{(\varepsilon + \phi M_r)^2 + M_r^2 + (\varepsilon/t_r)^2}{2\mu}, \\ U_- &= \frac{(\varepsilon + \varepsilon D_r)^2 + (\varepsilon/t_r)^2}{2\mu} + I\varepsilon D_r, \\ \Delta(s) &\le \min\{U_1, \max(U_+, U_-)\}. \end{aligned} \tag{S58}$$

At $r = 1$ the negative-m residual needs no bound on $|m|$: its slack and radial residual are exactly zero. At $r < 1/2$ only U_1 is used. These gap bounds use an upper bound on one physical curvature. Complete gradient recovery requires the signed derivative values specified in the main text.

C. UNIFORM CONSTANTS AND THE TOLERANCE EXPONENTS

Put $\delta = 1 - r$ and $K_0 = \bar{I}L$. The following bounds hold:

$$\begin{aligned} &t_r \ge 1, \quad \phi \le 1, \quad a_r \ge \delta && (0 \le r \le 1), \\ &a_r \le 2\delta, \quad k_r \ge 1 && (1/2 \le r \le 1), \\ &1 - \phi = \delta^2/(1+r^2), \quad D_r \le \delta. \end{aligned} \tag{S59}$$

For example, $a_r \ge \delta$ follows from

$$2(1+r) - (1+r^2)^2 = (1-r)(r^3 + r^2 + 3r + 1) \ge 0.$$

For $r \ge 1/2$, $a_r \le 2\delta$ follows from $(1+r^2)^2 - (1+r) = r(r^3 + 2r - 1) \ge 0$. Also

$$k_r - 1 = \frac{(1-r)(r^5 + r^4 + 4r^3 + 8r^2 + 11r - 1)}{(1+r^2)^3} \ge 0$$

on that interval. The bound on D_r is immediate; in fact $D_r \le \delta/2$ also holds, but is not needed for the stated constant.

Fix any $0 < h \le 1/2$. If $\delta \ge h$, first-order inversion gives $\|g\|^2 \le 3\varepsilon^2 + 3\varepsilon^2/a_r^2 \le 6\varepsilon^2/\delta^2$, so

$$\Delta(s) \le \frac{3\varepsilon^2}{\mu h^2}. \tag{S60}$$

If $\delta < h$, then $r > 1/2$ and (S56) gives $m_+ \le \nu + 4K_0\delta^2$. For $m \ge 0$, the zero-multiplier residual obeys

$$\|g\|^2 \le 3\varepsilon^2 + 3m^2 \le 3\varepsilon^2 + 6\nu^2 + 96K_0^2\delta^4. \tag{S61}$$

For $m < 0$ and $r < 1$, $|m|(1-\phi) \le \varepsilon D_r \le \varepsilon\delta$. With $\lambda = -m$, this gives $\|g + \lambda n\|^2 \le 5\varepsilon^2$ and $\lambda\sigma \le \bar{I}\varepsilon\delta$. At the exact

fold these claims hold because the corresponding slack terms vanish. Combining both signs,

$$\Delta(s) \le \frac{5\varepsilon^2 + 6\nu^2 + 96K_0^2h^4}{2\mu} + \bar{I}\varepsilon h. \tag{S62}$$

For $0 < \varepsilon \le K_0/8$, set $h = (\varepsilon/K_0)^{1/3} \le 1/2$. Let $X = K_0^{2/3}\varepsilon^{4/3}/\mu$. Then $\varepsilon^2/\mu \le X$ and $\bar{I}\varepsilon h \le X$, the latter because $\mu \le L$. The far bound is $3X$ and the near bound is at most $(5/2+48+1)X + 3\nu^2/\mu$. Thus 52 is a conservative common coefficient. When $\varepsilon = 0$, first-order inversion gives $g = 0$ away from the fold; at the fold the same multiplier argument yields the remaining ν^2 bound.

D. NOISY OBSERVATIONS AND NATIVE TOLERANCES

The main-text error inflation follows directly from the supplied absolute/norm budgets. In particular, $c_r \ge \widehat{c}_r - E_c \ge -I\nu$, including when the supplied curvature is positive. The proof applies to any valid supplied budgets. Establishing rounding-error budgets is a separate numerical task. Model constants and geometry must correspond to the objective actually optimized.

Suppose a native infinity-norm gradient test satisfies $\|\nabla_{a,w} f\|_\infty \le t_g$, with $t_g \ge 0$, and $I \ge I_{\min} > 0$, $|I'| \ge J_{\min} > 0$. Then

$$\begin{aligned}\varepsilon_{\text{actual}} &:= \max\{|b|, \|j\|_2/I\} \\ &\le \varepsilon_{\text{native}} := \max\{t_g/J_{\min}, \sqrt{2}\,t_g/I_{\min}\}.\end{aligned} \tag{S63}$$

Thus $\varepsilon = \varepsilon_{\text{native}}$ is a valid tolerance value for the theorem. Similarly, a scalar-curvature test $c_r \ge -t_H$, with $t_H \ge 0$, gives

$$\nu_{\text{actual}} := \max\{0, -c_r/I\} \le \nu_{\text{native}} := t_H/I_{\min}.$$

When curvature is required, use $\nu = \nu_{\text{native}}$. The other theorem assumptions, including $\varepsilon_{\text{native}} \le K_0/8$, must also hold. Supplied derivative errors require the error-inflation formulas in the main text. For an exponential intensity coordinate $J_{\min} = I_{\min}$. Native objective-reduction tolerances or successful line-search flags alone do not supply these derivative bounds.

For target gap τ, the two main-text thresholds allocate half of τ to each term. The example in the code has $\mu = L = 1/2, \bar{I} = 8, \tau = 10^{-6}$. Rounding its exact thresholds downward gives the sufficient choices $\varepsilon \le 2.886\times10^{-7}$ and $\nu \le 2.886\times10^{-4}$ in the stated normalized units. The pointwise bound can accept points rejected by the coarse uniform threshold.

E. SHARP EXAMPLES AND FAILURE OF UNRESTRICTED TRANSFER

For the first tolerance take the single objective $f(s) = \frac{1}{2}\|s - (1,e)\|^2$ with fixed unit e. Let $r \uparrow 1$, $\phi = 2r/(1+r^2)$, $I = (1+\phi)/(1+\phi^2)$. These intensities are bounded by two. Define $m = I\phi - 1 = (\phi-1)/(1+\phi^2)$. The intensity derivative satisfies $b = (I-1) + \phi m = 0$. Exact formulas are

$$\begin{aligned}\varepsilon = a_r|m| &= \frac{2(1-r)^3(1+r)}{(1+r^2)(r^4+6r^2+1)}, \\ \Delta(s) &= \frac{(1-r)^4}{2(r^4+6r^2+1)}.\end{aligned} \tag{S64}$$

Here $c_r/I = Ia_r^2 + \kappa_r m \ge 0$ for $0 < r < 1$: $Ia_r^2 \ge 0$, $m < 0$ and $\kappa_r < 0$. Hence $\nu = 0$, while $\varepsilon \sim (1-r)^3/4$ and $\Delta \sim (1-r)^4/16$. Any putative uniform upper bound $C\varepsilon^p$ with $p > 4/3$ and fixed C fails. The exact intensity adjustment is essential; keeping $I = 1$ would leave a first intensity derivative of order $(1-r)^2$ and would not establish this claim.

For the curvature tolerance let $s = (1,e)$, $n = (-1,e)$, $0 < \alpha < 1$, and $f_\alpha(u) = \frac{1}{2}\|u - z_\alpha\|^2$ with $z_\alpha = s - \alpha n$. The center $z_\alpha = (1+\alpha, (1-\alpha)e)$ is in $\mathcal{C}$. At the fold, $g = \alpha n$, $b = 0, j = 0, c_r = -\alpha$, and $\Delta = \alpha^2$. Thus the ν^2 order cannot be improved. This is the Euclidean-metric version of the fold ambiguity family used earlier. It also proves that first-order tolerances alone cannot imply a vanishing gap at the fold.

For unbounded outer coordinates, use the fixed objective $f(s) = \frac{1}{2}\|s - (1,1/2,0)\|^2$. Let $r \to \infty$, $e = (1,0)$ and $I = (1+\phi/2)/(1+\phi^2)$. Again $b = 0$, while $m = I\phi - 1/2 \to -1/2$, $a_r \sim -2/r^2$, and $\kappa_r \sim 4/r^3$. Therefore

$$\varepsilon \sim r^{-2}, \qquad \nu \sim 2r^{-3}, \qquad \Delta(s) \to 1/8. \tag{S65}$$

No bound depending only on vanishing ε, ν can hold over both unrestricted sheets. Canonicalization changes the derivatives: the inner first derivative in this example does not remain small.

For intensity scaling, set $w = 0$, $I = \exp a \downarrow 0$ and $f(s) = \frac{1}{2}\|s - (1,0,0)\|^2$. The raw intensity derivative is $I(I-1)$, its second derivative is $I(2I-1)$, and the polarization Hessian is $4I^2\,\mathrm{Id}_2$ with zero mixed block. Raw first derivatives and the negative part of the full raw Hessian tend to zero, but $\Delta \to 1/2$. In contrast, $b = I - 1$ does not vanish, so the normalized test rejects the sequence. This explains why a lower intensity/coordinate-derivative bound is needed when converting a raw solver tolerance.

F. RELATION TO SQUARED SLACK AND KL RESULTS

The scalar slack of the normalized polarization boundary is

$$I - \|p\| = I\frac{(1-r)^2}{1+r^2} = \left(\frac{\sqrt{I}(1-r)}{\sqrt{1+r^2}}\right)^2. \tag{S66}$$

Locally at a nonzero fold this is a squared slack in a regular radial coordinate. This identity connects the fold to squared-variable theory. The proof combines the corresponding chain-rule geometry with the standard dual-residual bound. Ding–Wright [7, Thm. 3.6, Cor. 3.7] already establish quantitative approximate second-order transfer with active-Jacobian rank conditions and conditioning-dependent constants. Their conclusion is an approximate weak second-order condition, whereas the present strongly convex specialization reports an explicit gap from a single radial lower test.

Ouyang *et al.* [8, Thm. 3.11, Rem. 3.12] establish local KL exponents for Hadamard difference models under a KL assumption and an additional error bound, including exponent $3/4$ without strict complementarity. A gradient-to-gap power $4/3$ is therefore a known type of degenerate-contact behavior. The Stokes result makes this power explicit on the stated coordinate domain, with uniform constants, a scalar curvature test, and a supplied-error model. Li *et al.* [9, Thm. 5,

Lem. 6.1] provide a separate algorithmic guarantee under nondegeneracy/strict-saddle assumptions and qualitative proximity of approximate stationary sets. The main-text theorem uses stronger convexity of the physical objective, but does not require those assumptions or a known neighborhood of its optimum. These comparisons delimit the specialization addressed here.

For a general nonconvex C^2 objective, the existing recovery-error bound can instead be combined with a standard projected-gradient residual. With the Euclidean projection $\Pi_{\mathcal{C}}$ in (S31), define $R_\gamma(g) = \gamma^{-1}\|s - \Pi_{\mathcal{C}}(s - \gamma g)\|$ for $\gamma > 0$. Euclidean nonexpansiveness gives $R_\gamma(g) \le R_\gamma(\widehat{g}) + \|g - \widehat{g}\|$. This standard inequality gives an approximate stationarity statement for the nonconvex case.

G. FIXED VERIFICATION AND REPRODUCTION

The fixed grid has three SPD metrics, intensities $0.02, 1, 4$, radii $0, 0.25, 0.5, 0.75, 0.9, 0.99, 0.999, 1$, two directions and four gradient-center choices, giving 576 quadratics. Twenty-four additional strongly convex quartics use the ideal metric plus $0.1(v^{\mathsf{T}}(s - z))^4/4$, $v = (1, 0.3, -0.2)$, with the same intensities, four radii and two directions. Their Hessian's actual polarization upper eigenvalue supplies a valid pointwise L. Supplied derivative perturbations have budgets zero, 10^{-8} or 10^{-4}. Every result is retained; no configuration is added in response to its outcome. SLSQP supplies independent physical minima, with 600 native successes and maximum feasibility violation 9.64×10^{-14}. Computed gap-minus-bound residuals reach 1.24×10^{-14}, consistent with that reference accuracy; these are not symbolic proofs. Independent PyTorch derivatives check all 600 objectives, including the prescribed radial directions at the origin, where the coordinate alone does not encode that direction.

Ninety-digit decimal calculations retain 12 first-tolerance and 12 curvature-tolerance examples, eight unbounded-outer examples and 16 vanishing-intensity examples. The last first-tolerance ratio is 0.3968502629915207, approaching $2^{-4/3} = 0.3968502629920499$. The constructions and their limiting calculations above provide the proofs; finite precision checks do not establish the exponents on their own.

The retrospective endpoint check uses all 9 284 previously stored real-data warm endpoints. Exactly 463 are switched inward, preserving their outputs. Derivative tests are recomputed at the inner coordinates; $\mu = L = 1/2$, $\bar{I} = 8$ and zero supplied derivative-error budgets are used. All recorded intensities satisfy that domain. The pointwise bound accepts 8 861 endpoints and rejects all 295 historical false successes, plus 128 accurate endpoints. The coarse uniform bound alone accepts 980. This check makes no new optimizer calls but acquires 9 284 full head Hessians, including points where the theorem would not require one. The implementation prioritizes complete coverage of the stored endpoints over monitor cost. Validated rounding-error budgets and evaluation on unseen observations remain outside this retrospective check.

Sources, fixed protocols, all numeric records, a standalone certificate implementation, an independent AD audit and replay scripts are in theory/stopping_guarantee/. Prior experiments and their files remain unchanged.

S21. OUTWARD EVALUATION OF THE NOMINAL STOPPING BOUND

Section S20 gives bounds in real arithmetic. Here we evaluate its pointwise expression with outward intervals for the nominal ideal four-analyzer quadratic. This adds numerical enclosure of the intensity, coordinate geometry, and derivative tests. The model remains $f(s) = \frac{1}{2}\|As - y\|^2$, with the ideal A in Section S1 and $\mu = L = 1/2$. Each stored binary64 coordinate and observation denotes its exact real value. The quantity certified is the optimization gap for that model and those inputs.

A. ARITHMETIC AND INPUT SEMANTICS

The interval layer follows the standard inclusion principle [10, Secs. 5.3–5.5]. For p fractional bits, endpoints are integers divided by $Q_p = 2^p$. Additions and subtractions use exact integer arithmetic. Multiplication and division round the lower endpoint downward and the upper endpoint upward using integer floor and ceiling division. Division by an interval containing zero returns an inconclusive result. Square roots use integer square roots of the scaled endpoints, increasing the upper root by one exactly when it is not a perfect square.

The exponential uses a separate enclosure rather than a library approximation. For an endpoint $x \in [-64, 64]$, range reduction gives $t = |x|/2^k \le 1/8$. With $N = 48$, positivity and a geometric bound on the remaining term ratios give

$$\begin{aligned} S_N(t) &= \sum_{j=0}^{N} \frac{t^j}{j!}, \\ S_N(t) \le e^t &\le S_N(t) + \frac{t^{N+1}}{(N+1)!}\frac{1}{1 - t/(N+2)}. \end{aligned} \tag{S67}$$

The sum and remainder are enclosed by integer interval operations. Repeated squaring reverses the range reduction; reciprocation handles negative endpoints. Monotonicity supplies the enclosure for an interval argument, including one that crosses zero. Arguments outside the supported range return inconclusive. All operations determining acceptance use integers; conversion to a decimal or binary64 display occurs afterward.

For an outer input, the implementation uses the exact inner preimage $C(w) = w/\|w\|^2$ algebraically. It evaluates the physical state from the original coordinates, and the derivative tests at this inner preimage. The identity $F(C(w)) = F(w)$ preserves the exact physical output. The original squared radius is compared with one using exact rational arithmetic. Thus the inner radius is known to lie in $[0, 1]$, and intersecting its computed enclosure with this interval is justified by the map identity. The same reasoning permits the nonnegative support-slack ratio in Section S20. No rounded optimizer restart is introduced.

TABLE S4. Outward pointwise stopping with the indicated number of fractional bits on the same 9 284 endpoints. "Accurate rejected" includes inconclusive results. All four precisions accept zero inaccurate endpoints. Times describe one CPU pass; the separate cone-reference calculation takes 7.69 s.

Bits	Certified	Inconcl.	Accurate rej.	Time (s)
32	8 858	3	131	6.21
64	8 861	0	128	6.38
128	8 861	0	128	7.32
256	8 861	0	128	8.73

B. CORRECTNESS OF AN ACCEPTED BOUND

Proposition S8 (Validated nominal stopping)**.** *For finite binary inputs (a, w, y), suppose the supplied interval monitor with target tolerance 10^{-6} reports a certified result. Then the mathematical mapped state $s = T_F(a, w)$ satisfies*

$$f(s) - \min_{u \in \mathcal{C}} f(u) \le 10^{-6}. \tag{S68}$$

Proof. The input conversion encloses the exact binary values. Each elementary operation preserves containment by outward integer rounding. Equation (S67) encloses $I = e^a$; a successful evaluation also verifies $0 < I \le 8$. Exact inner-preimage selection gives the domain of the Section S20 pointwise bound. For the ideal quadratic, $e^{\mathsf{T}} H_{pp} e = \mu = 1/2$ exactly.

The implementation encloses b, $\|j\|/I$, and c_r/I, and hence the sufficient ε and ν. It evaluates the first-order and one-sided-curvature expressions in (S55) and (S58) with intervals, including all geometry operations. Minima and maxima preserve containment of the corresponding real values. At the origin, $DF = 2\,\mathrm{Id}_2$ supplies the first-order expression directly. Every accepted bound therefore encloses a valid upper bound on the real gap. If its upper endpoint is U/Q_p, the final comparison is the exact integer test $10^6 U \le Q_p$, proving (S68). An unresolved denominator or domain check returns inconclusive rather than accepting. □

The implementation evaluates a single scalar radial curvature when $r \ge 1/2$. It obtains the derivative enclosures analytically from the nominal model and forms no full head Hessian. This model-specific route also has access to physical gradients; the direct physical controls remain applicable.

C. RETROSPECTIVE VERIFICATION AND PRECISION

The protocol fixes fractional precisions $32, 64, 128, 256$ and all 9 284 stored warm endpoints before this arithmetic comparison. States, observations, histories, and optimizer results are unchanged. A separate outward implementation of the direct cone reduction at 256 bits classifies 8 989 endpoints below the gap threshold and 295 above it, with no threshold-overlap interval. Each accepted stopping result is checked against this independent cone expression.

Table S4 shows identical decisions at 64–256 fractional bits and agreement with the earlier float64 pointwise decisions. All 295 inaccurate endpoints are rejected; 128 accurate endpoints also remain rejected. At 32 bits, three additional accurate endpoints are inconclusive because their radius enclosures include zero in a required denominator. Their true inner radii are approximately 9.84×10^{-6}, 2.05×10^{-6}, and 6.67×10^{-6}. They are retained in the denominator and the reported rejection count.

At every precision, the monitor makes 52 scalar curvature evaluations and forms zero full head Hessians. The previous coverage implementation acquired 9 284 full head Hessians. The new accounting concerns this analytic nominal-model implementation; Table S4 reports its interval cost separately.

Independent exact-rational checks verify the arithmetic, using a direct Taylor-series exponential oracle at the original argument rather than the implementation's range reduction. Additional Cartesian Jacobian/Hessian checks at exact $I = 1$ verify 196 radial-tolerance enclosures on 49 inputs. The frozen stress suite has 43 inputs and 172 precision-specific checks, including exact folds, nearby inner and outer states, low intensity, and unsupported inputs. Unsupported inputs remain inconclusive. The mathematical enclosure argument above supplies the acceptance implication; these checks test its implementation.

D. REPRODUCTION

The implementation requires only the Python standard library. From the source-package root, run `theory/validated_stopping/replay.py` with `--output` naming a new directory. The script reproduces the records and compares their integer endpoints exactly. A minimal folded optimum for cli.py uses raw coordinates $(0, 1, 0)$ and observations $(1, 1/2, 0, 1/2)$. The README gives complete one-line commands.

The source protocol, integer bound endpoints, independent checks, all four precision runs, and replay comparisons are retained in theory/validated_stopping/. Applying the method to uncertain instrument parameters requires enclosures for that expanded physical model.

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